



In-depth Analysis for Lanaken

Leveraging Data Analytics for Efficient OR Planning at ZOL

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Introduction

Efficient management of operating rooms (ORs) requires an understanding of how planned surgical activity aligns with real-world execution. This document provides an in-depth, site-specific analysis for **Campus Lanaken**, serving as an analytical extension of the previous exploratory study. The goal is to systematically address the key research questions formulated in the project description:

Research questions

- **Planned versus observed timing**
 - How well do the planned operating durations correspond to the actual durations?
 - What is the typical deviation between the planned start time and the actual entry into the operating room?
 - Which specialisms or procedure types show the largest differences?
 - How do these deviations scale relative to the planned duration (proportional or percentage-wise)?
- **Adherence to working hours**
 - How often do procedures run beyond regular working hours (08:00 to 16:30)?
 - What proportion of OR activity occurs in the evening or at night?
 - How do these patterns differ between campuses?
- **Room allocation**
 - How often are procedures performed in a different room than originally planned?
 - Are these reallocations concentrated in specific rooms or specialisms?
- **Urgency and scheduling**
 - How are urgent and non-elective cases distributed across campuses and time periods?
 - To what extent do urgent surgeries overlap with elective activity and cause scheduling disruptions?

Each of the following sections addresses one of these themes in depth, focusing on the local characteristics of the Lanaken campus.

Unique values and data richness

The dataset captures a high degree of heterogeneity in surgical practice. A total of 34889 unique patient IDs are recorded, spread across 588 distinct procedure codes and 578 procedure names. Surgeon and anesthesiologist coverage is broad, with 97 and 132 unique identifiers respectively. These numbers reflect the diversity of clinical practice at ZOL and underscore the importance of developing grouping strategies to obtain actionable insights. Table 1 provides an overview of the number of unique values per variable.

Table 1: Number of unique values per variable.

Variable	UniqueValues
PatientID	34889
ProcedureCode	588
ProcedureName	578
AnesthesiologistID	132
HeadSurgeonID	97
LengthOfStay	66
PlannedOR	9
ActualOR	7
Weekday	5
Year	4
AdmissionType	3

Surgical procedure durations

Operating time is one of the most critical variables for planning. Procedure durations display wide variation, both within and across procedure types. Percentile analysis shows that 95% of all procedures finish within 86 minutes, 98% within 114 minutes, 99% within 133 minutes, and 99.9% within 196 minutes. While the majority of procedures are considerably shorter, the long right hand tail in the distribution indicates the presence of outliers and highly complex cases that drive variability in scheduling. Figure 1 illustrates the distribution of procedure durations, and Table 2 summarizes the percentile values.

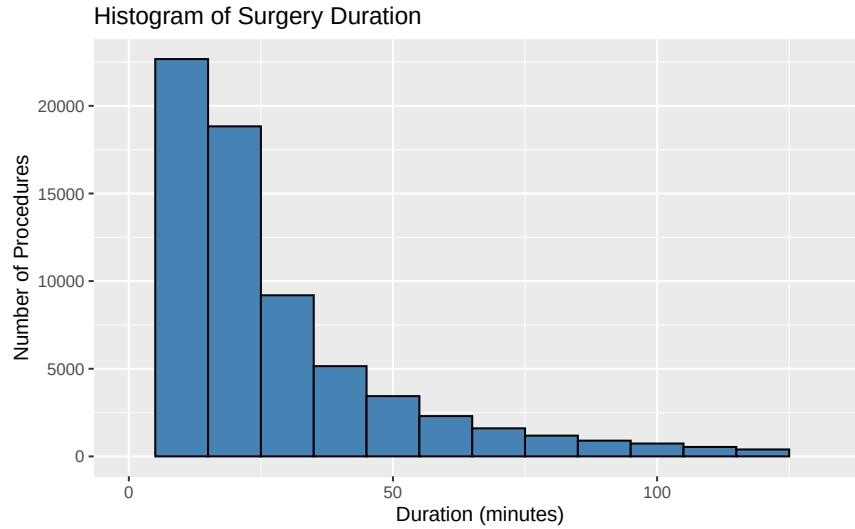


Figure 1: Histogram of Surgery Duration

Table 2: Percentiles of surgical procedure duration (in minutes).

Percentile	Duration (minutes)
95%	86
98%	114
99%	133
99.9%	196

Looking at the most frequent procedures illustrates the large variation in operating times. FACO EMULSIFICATIE has an average duration of 27.6 minutes and a median of 26 minutes, and transforaminele epidurale infiltratie : lumbaal averages 12.6 minutes with a median of 12 minutes. Facet radiofrequency lumbaal typically lasts 18.8 minutes on average with a median of 18 minutes. In contrast, VITRECTOMIE have an average duration of 55.3 minutes and a median of 53 minutes, highlighting the substantial difference in resource requirements between routine short procedures and major surgical interventions. This mix of short standardized cases and longer complex operations underscores the operational challenge of planning an efficient operating room schedule that accommodates both types of activity. Table 3 summarizes the ten most frequent procedure types.

Table 3: Top 10 most frequent procedure types with mean and median duration

ProcedureName	Count	Mean	Median
FACO EMULSIFICATIE	11025	27.6	26
transforaminele epidurale infiltratie : lumbaal	7186	12.6	12
facet radiofrequency lumbaal	5577	18.8	18
facet diagnostisch lumbaal	4962	13.7	13
proefsacrale infiltratie	3290	11.5	11
transforaminele epidurale infiltratie : sacraal	2363	12.3	11
epidurale infiltratie : cervicaal	2091	14.0	13
VITRECTOMIE	1788	55.3	53
facet radiofrequency cervicaal	1693	22.8	22
facet diagnostisch cervicaal	1555	16.6	16

Admission type and urgency

The majority of procedures in the dataset are performed in a planned setting. Ambulatory surgery accounts for 98.1%, and inpatient hospitalization for 1.9%, while emergency admissions represent only one recorded case. The urgency classification shows a similar pattern: 99.6% of procedures are elective, and 0.4% are non elective. Although relatively small in number, the latter category remains important from a planning perspective, as urgent procedures may disrupt elective schedules and lead to cancellations or overtime. Table 4 and Table 5 summarize both distributions.

Table 4: Frequency table for Admission Type

AdmissionType	Count	Percentage
DAG	68108	98.1
HOS	1286	1.9
SPOED	1	0.0

Table 5: Frequency table for Urgency Classification

UrgencyType	Count	Percentage
electief	69109	99.6
niet-electief	286	0.4

Temporal distribution of activity

Surgical activity is spread consistently across weekdays. Fridays account for 18.5 percent of weekly activity, Tuesdays for 21.5 percent, Thursdays for 19.1 percent, Wednesdays each for 19.8 percent, and Mondays each for 21.2 percent. There are no Weekend days shown in the data. Annual trends reveal steady growth, with the number of procedures increasing from 20016 in 2022 to 20636 in 2024. At the time of data extraction, 8702 procedures in 2025 represented 12.5 percent of the total. Together, Table 6 and Table 7 summarize the distribution of surgical activity across weekdays and years.

Table 6: Frequency table for Day of the Week (sorted by descending count)

Weekday	Count	Percentage
Di	14902	21.5
Ma	14710	21.2
Wo	13742	19.8
Do	13233	19.1
Vr	12808	18.5

Table 7: Frequency table for Year

Year	Count	Percentage
2022	20016	28.8
2023	20041	28.9
2024	20636	29.7
2025	8702	12.5

In summary, procedure durations show strong variability with a long tailed distribution driven by complex surgeries. The majority of procedures are elective, while non elective cases still have an important impact on schedule stability. Activity is distributed evenly across weekdays and has grown consistently in recent years. These findings form the baseline against which more advanced performance metrics such as start and end deviations, utilization rates, and predictive models can be developed in subsequent sections of this report.

How accurately does planning reflect reality in the OR?

Are planned and observed durations aligned?

The comparison between planned and observed surgery durations shows a broad spread around the one to one line. Many procedures lie close to the diagonal, indicating that their observed durations correspond reasonably well to the planned values. At the same time, a substantial number of points fall above or below this line, showing that both underestimation and overestimation occur frequently. Most deviations appear as longer than planned surgeries, visible in the dense area below the red dashed line. The dispersion increases for longer procedures, reflecting greater uncertainty for complex cases. Figure 2 displays this relationship.

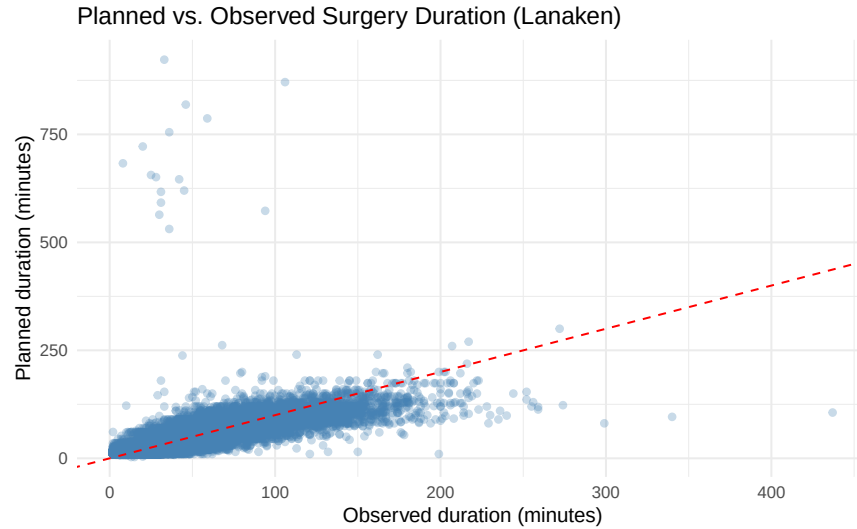


Figure 2: Planned vs. Observed Surgery Duration (Lanaken)

On average, surgeries at Lanaken show meaningful variation between planned and actual durations. Among procedures that last longer than planned, the mean deviation is 9 minutes with a median of 5 minutes, while shorter cases deviate on average 7.1 minutes with a median of 5 minutes. The standard deviations of 11.2 minutes and 15.2 minutes highlight the presence of both moderate deviations and substantial outliers. While most differences remain within reasonable limits, a small number of extreme values extend the overall range.

Almost half of all surgeries (41.7%) take longer than planned, with an interquartile range of 8 minutes, while 58.3% finish earlier, with an interquartile range of 7 minutes. This distribution shows that deviations occur in both directions, but the spread is larger for cases that finish earlier. The considerable variability within each category indicates that although overall planning accuracy is reasonable at the aggregate level, individual durations remain difficult to predict. Improving the consistency of duration estimates, particularly for procedures that frequently deviate, could help reduce daily schedule variability and limit delays across the operating list. Table 8 provides a detailed overview of these differences.

Table 8: Difference between actual and planned duration (minutes), separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99_9	Percent
Longer than planned	9	5	11.2	8	1	331	29	53.5	108	41.7%
Shorter than planned	7.1	5	15.2	7	0	890	21	40	80.1	58.3%

Expressing deviations as a percentage of the planned duration provides a standardized view across short and long procedures. Most cases fall within about ± 40 percent of the planned time, indicating a generally acceptable level of scheduling accuracy in routine practice. The histogram peaks slightly below zero, showing that surgeries are on average completed a bit earlier than planned. The shape is roughly symmetrical, although the right tail extends further, indicating that large overruns occur less frequently but can be substantial when they appear. This pattern suggests that while most procedures are reasonably well estimated, a smaller group of cases contributes disproportionately to the overall variability in duration accuracy. Figure 3 shows the full distribution.

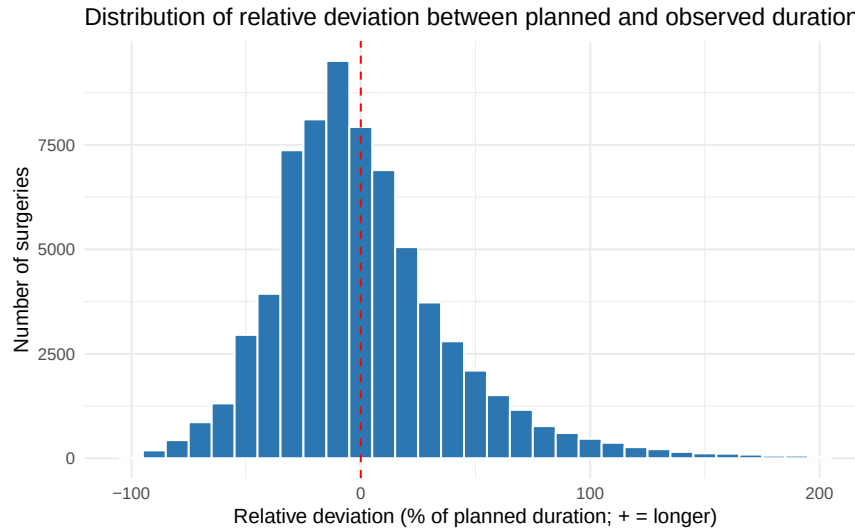


Figure 3: Distribution of relative deviation between planned and observed duration

Duration deviations differ markedly between admission types. Inpatient, HOS, cases show the highest variability, with an average deviation of 14.5 percent and a median of 6.7 percent, while the coefficient of variation is 3.4. This means that even if the mean deviation seems moderate, individual cases differ strongly from their planned duration, which is typical for unpredictable acute interventions.

Inpatient, SPOED, surgery has a deviation of -17.8 percent and a median of -17.8 percent, suggesting that this case was a rare occurrence. The coefficient of variation cannot be calculated as there is only one recorded case of an emergency surgery.

Ambulatory, DAG, surgeries are the most predictable group. Their mean deviation is 2 percent, the median is minus 5 percent, and the coefficient of variation of 21.9 mainly reflects the fact that even moderate absolute differences translate into larger percentages for short procedures. These results show that standardized short-stay procedures are generally estimated accurately in the planning process. Table 9 summarizes these patterns.

Table 9: Relative duration deviation and coefficient of variation by admission type (Lanaken)

AdmissionType	n	Mean relative deviation (%)	Median relative deviation (%)	SD (%)	CV (%)
HOS	1286	14.5	6.7	49	3.4
DAG	68108	2	-5	42.9	21.9
SPOED	1	-17.8	-17.8	NA	NA

A similar pattern appears when comparing elective and non elective surgeries. Non elective procedures have an average relative deviation of -2.6 percent and a median of minus 7.3 percent, with a coefficient of variation

of -12.1, highlighting their unpredictable character. Elective operations show smaller deviations, with an average of 2.2 percent and a median of minus 2.2 percent. Their coefficient of variation of 19.5 results from the influence of a small number of extreme outliers rather than consistent misestimation. Overall, these findings confirm that while elective cases generally follow expected durations well, non elective surgeries remain the main source of variability in the operating schedule. Table 10 summarizes these results.

Table 10: Relative duration deviation and coefficient of variation by urgency type (Lanaken)

UrgencyType	n	Mean relative deviation (%)	Median relative deviation (%)	SD (%)	CV (%)
electief	69109	2.2	-4.9	43	19.5
niet-electief	286	-2.6	-7.3	31.4	-12.1

The mean relative deviation between planned and observed durations remains broadly stable over the observed years. In 2022 the average deviation is 0.1 percent, increasing slightly to 1.9 percent in 2023. A clearer rise appears in 2024, reaching 4.8 percent, while the partial data for 2025 show a decrease to 2.1 percent.

This pattern indicates that planning accuracy at Lanaken has remained relatively consistent over time, with only moderate year to year fluctuations. The differences are small and likely reflect normal operational variation rather than systematic changes in planning quality. Overall, the stability of the yearly averages suggests that the underlying scheduling process is well established, even though individual level variability continues to affect day to day accuracy. Figure 4 illustrates these trends.

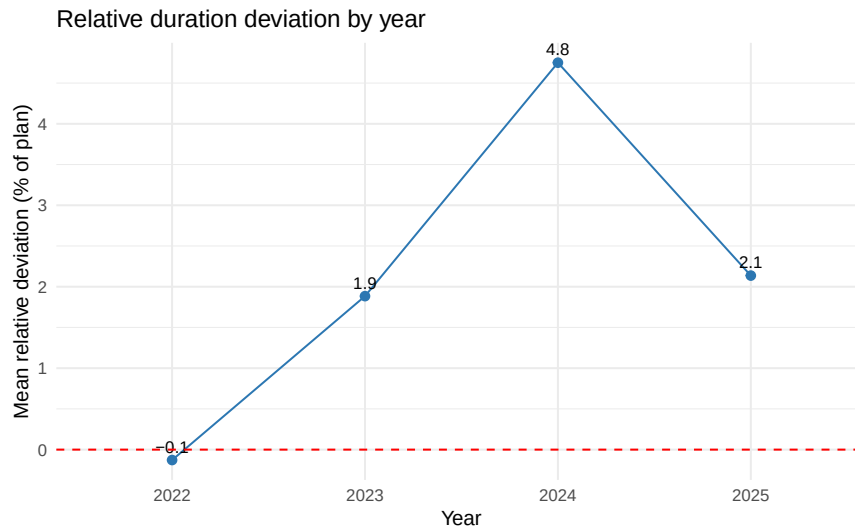


Figure 4: Relative duration deviation by year

The relative deviation between planned and observed durations at Campus Lanaken shows a consistent pattern across the regular workweek. Mean deviations range from 0.8 percent to 3.9 percent, indicating that planning accuracy is generally stable from Monday through Friday. The slightly higher deviation on Tuesday, followed by a moderate value on Wednesday, likely reflects variation in the types of procedures performed rather than structural scheduling differences. Lower values on Thursday and Friday suggest that case mix later in the week may be more predictable. Overall, the figure shows that at Lanaken the alignment between planned and observed durations is steady across weekdays, with only modest fluctuations tied to day specific activity profiles.

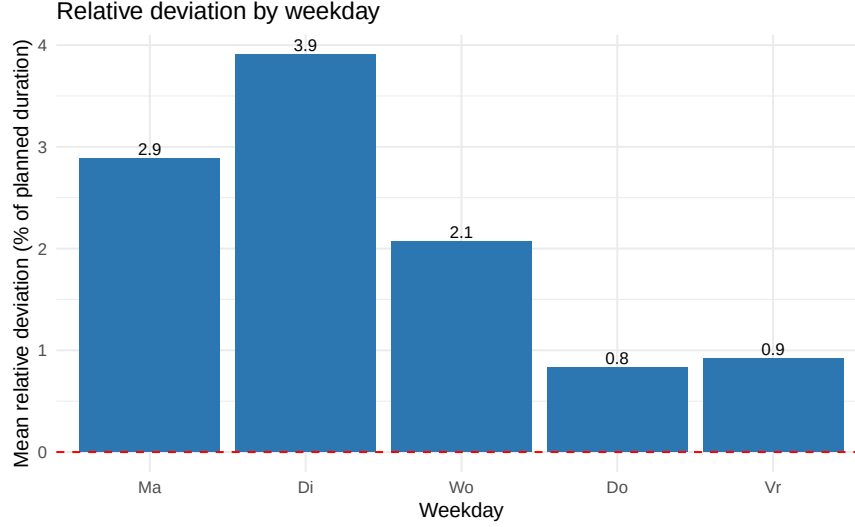


Figure 5: Relative deviation by weekday

Variation in deviation differs considerably across operating rooms. The largest average deviations are concentrated in the complex surgery rooms of the LO block, most notably LO03, where the mean absolute deviation is 16.4 minutes and 45.9 percent of cases run longer than planned. Other high variability rooms such as LO05, LO01, and LO04 also show average deviations between 10.6 and 13.5 minutes, reflecting the higher uncertainty typical of lengthy or technically demanding procedures.

By contrast, rooms such as LP01 and LP02, mainly associated with short and standardized interventions, display mean deviations below 6 minutes and a roughly balanced share of shorter and longer cases. These differences illustrate how the degree of procedural complexity and specialization strongly influences planning precision. Table 11 provides the detailed overview.

Table 11: Duration deviations per operating room, ordered by mean absolute deviation

ActualOR	n	Mean abs dev	Longer (%)	Mean relative dev (longer)	Shorter (%)	Mean relative dev (shorter)
LO03	4196	16.4	45.9	29.3	51.1	-21.2
LO05	5206	13.5	41.2	30.8	55.2	-23.4
LO01	5202	12.3	39.2	27.6	57.5	-21
LO04	6009	10.6	45.1	30.7	50.8	-20.9
LO02	9974	7.8	41.5	30.1	52.9	-20.7
LP02	18619	5.6	38.6	43.7	54.8	-31.2
LP01	20189	4.8	43.6	46.3	48.8	-29.5

Among the most extreme deviations in surgery duration, longer and shorter cases occur in almost equal proportions. Around 66.8 percent of the top one percent of deviations correspond to procedures that last longer than planned, while 33.2 percent finish earlier than expected. This contrasts with earlier observations based on a different dataset, where overruns dominated more clearly.

Although both directions are represented among the extreme cases, their operational impact remains different. Early finishes generally create limited idle time, whereas overruns can delay subsequent cases and increase the likelihood of overtime. The near balance between longer and shorter cases in the current extreme deviations suggests that large underestimations and overestimations both contribute meaningfully to

schedule variability, reinforcing the need to monitor procedures prone to substantial deviations in either direction. Table 12 summarizes these results.

Table 12: Direction of deviation among the 1% most extreme cases (Lanaken)

Direction	n	Percentage
Longer than planned	474	66.8%
Shorter than planned	236	33.2%

A closer look on the coefficient of variation

The monthly coefficient of variation for planning deviation displays a highly volatile pattern, as shown in Figure 6. Most months cluster around relatively low values between 1.0 and 1.5, but these stable periods are repeatedly interrupted by sharp, isolated spikes. Early 2022 shows modest variation until a sudden jump to nearly 4, followed shortly by another month above 3. Similar single-month surges reappear throughout 2023 and 2024, with values that briefly exceed 3 before dropping back to the lower baseline.

These spikes are sporadic rather than sustained. They do not form a rising or falling trend but instead appear as short bursts of unusually high proportional deviation, likely driven by temporary shifts in case mix or isolated outliers. By contrast, 2025 exhibits much steadier behaviour, with the coefficient remaining close to 1.2 and no visible extreme peaks.

Overall, the pattern is characterised by a low, stable baseline punctuated by occasional large fluctuations. This indicates that planning accuracy at Lanaken is generally consistent, but certain months experience markedly higher relative dispersion, reflecting short-term variability rather than a structural change over time.

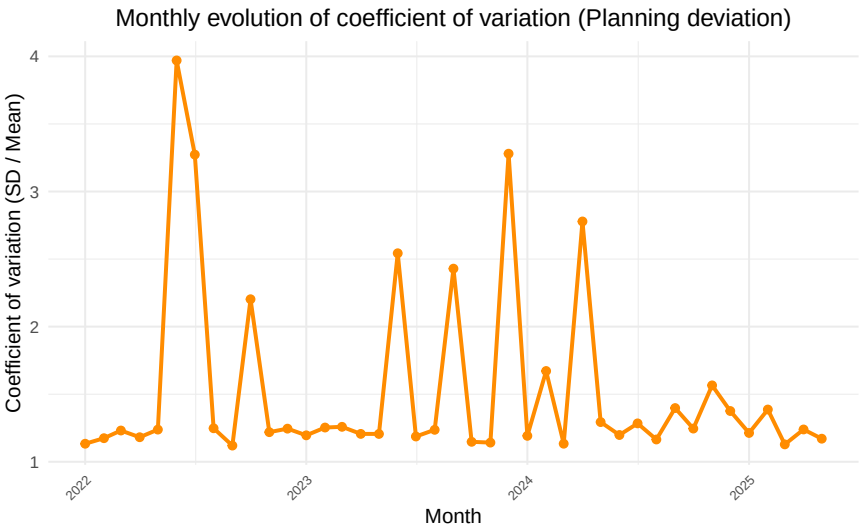


Figure 6: Monthly evolution of coefficient of variation (Planning deviation)

Variability by planned duration group

Variation in surgical timing can be examined from two complementary perspectives that describe different aspects of performance. The first is the coefficient of variation of observed duration, calculated as the standard deviation divided by the mean of the observed surgical times. This indicator expresses how dispersed

Table 13: Summary of mean duration, SD, and CV by planned duration group (Observed vs. Planning deviation)

Group	n	Mean duration	SD duration	CV duration	Mean planning	SD planning	CV planning
<30	50060	17.78666	9.267356	0.5210286	5.337355	5.284281	0.9900561
31–60	12315	43.15436	16.483787	0.3819727	10.223386	9.331324	0.9127430
61–90	4320	75.02523	25.877444	0.3449165	18.060185	15.146979	0.8386945
91–180	2662	108.92186	34.644410	0.3180666	23.638618	21.219601	0.8976667
> 180	38	109.57895	75.841810	0.6921203	307.578947	304.498936	0.9899863

the recorded durations are relative to their average. In a dataset such as this one, which contains a large number of both short and long operations, a high coefficient of variation mainly reflects the broad spread in case duration across procedure types rather than instability within individual surgeries.

The second perspective is the coefficient of variation of the planning deviation, which measures the consistency of planning accuracy. It is computed as the standard deviation of the absolute deviation between planned and observed time, divided by the mean of that deviation. This metric focuses on how variable the alignment between planned and realised durations is, regardless of how long the surgeries themselves last.

Comparing both perspectives helps disentangle two components of variability: the structural heterogeneity in procedure duration and the variability in how precisely these durations are planned. Understanding how these two aspects evolve provides insight into whether deviations primarily originate from the diversity of surgical activity or from variation in scheduling accuracy. The following table summarises these indicators across five duration categories and serves as a baseline for the more detailed, category-specific analyses presented in the subsequent subsections. All values are reported in Table 13, which provides the exact CV measures for both observed durations and planning deviations.

The summary statistics confirm a clear scaling effect in relative variability. The coefficient of variation of observed durations decreases as planned duration increases: procedures planned for less than 30 minutes have a CV of 0.5210286, procedures in the 31–60 minute group show 0.3819727, the 61–90 minute group 0.3449165, the 91–180 minute group 0.3180666, and the longest category over 180 minutes shows 0.6921203. This pattern is consistent with the fact that shorter cases amplify small absolute fluctuations when expressed relative to their mean, while longer cases distribute absolute variation over a broader time base.

For the planning deviation, coefficients of variation are considerably higher across all groups, reflecting the larger proportional spread in the differences between planned and actual duration. These values range from 0.9900561 in the shortest category to 0.9127430 in the 31–60 minute group, 0.8386945 in the 61–90 minute group, 0.8976667 in the 91–180 minute group, and 0.9899863 in procedures lasting more than 180 minutes. This pattern suggests that longer and more complex procedures, while not necessarily less stable in execution, exhibit greater variation in planning accuracy.

Together, these figures show that relative variability in observed duration is largely structural and reflects case mix composition, whereas variability in planning deviation captures the irregularity in scheduling precision. The distinction between the two metrics provides the foundation for the next analyses, which explore how these patterns manifest within each duration category and across operating rooms.

The monthly evolution of the coefficient of variation for observed durations shows a highly stable pattern for four of the five planned duration groups, as visible in Figure 7. The <30, 31–60, 61–90, and 91–180 minute groups all form dense, nearly flat trajectories over time, with only minor month-to-month fluctuations. These curves remain tightly clustered: the <30 minute group sits consistently around 0.4 to 0.5, whereas the three mid-length groups remain lower, typically between 0.25 and 0.4. The clear separation between short procedures and the mid-range groups persists throughout the entire period, reflecting stable proportional variability.

In contrast, the >180 minute group behaves very differently. Instead of a continuous line, it appears only sporadically and shows extreme swings, from near zero to values close to 1.0. These abrupt jumps reflect the

small number of very long procedures performed each month, where even a few cases can cause large shifts in the coefficient of variation. The volatility therefore stems from data sparsity rather than a structural change in duration variability.

Overall, the figure reveals a stable and well-behaved pattern for the majority of procedure lengths, while the longest-duration category displays irregular and highly sensitive behaviour due to low monthly counts.

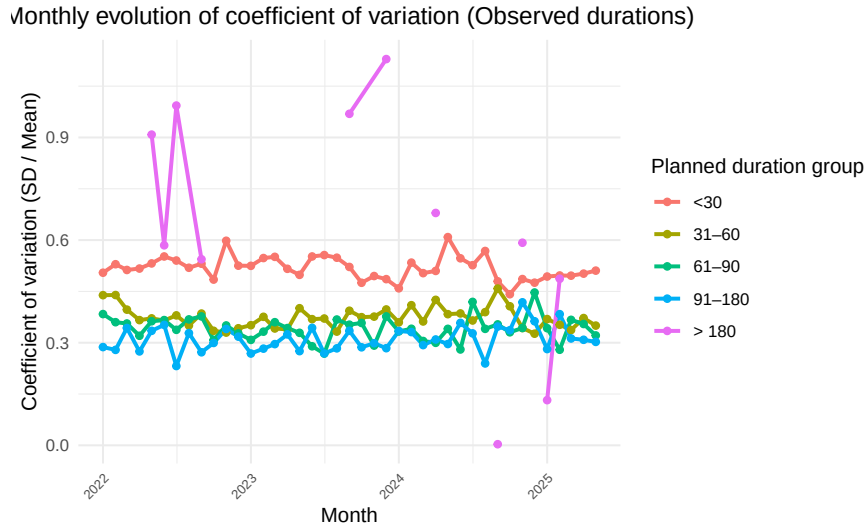


Figure 7: Monthly evolution of coefficient of variation (Observed durations)

The coefficient of variation of planning deviation shows a dynamic pattern over time, with marked differences between duration groups, as illustrated in Figure 8. Across most months, the mid-range categories (31–60, 61–90, and 91–180 minutes) form a compact cluster, fluctuating between roughly 0.7 and 1.1. Their tight trajectories indicate that planning deviations for these procedures are comparatively stable and change only modestly from month to month.

Short procedures (<30 minutes) display noticeably larger swings. Their curve oscillates more widely than the mid-range groups and reaches several pronounced peaks around 1.2 to 1.3. These spikes suggest periods in which small absolute deviations become proportionally large for very short cases, making the coefficient of variation more sensitive to case mix shifts.

The >180 minute group behaves quite differently: instead of a continuous trajectory, it appears intermittently with large jumps, sometimes falling below 0.6 and at other times rising above 1.3. These swings are driven by small monthly sample sizes, where just a few unusually short or long deviations can cause sharp changes in the coefficient.

Overall, the figure shows that planning deviation is most stable in the mid-duration range, more volatile for short cases, and highly irregular for the longest procedures because of sparse data.

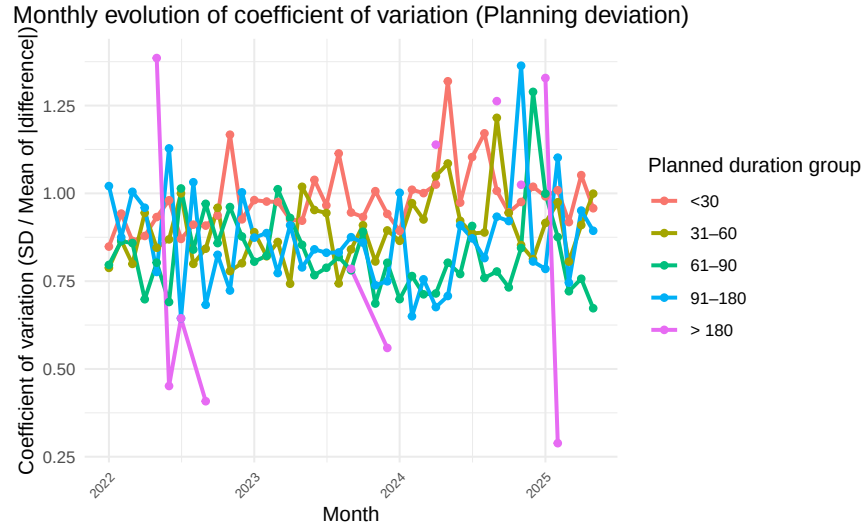


Figure 8: Monthly evolution of coefficient of variation (Planning deviation)

Together, these patterns indicate that the surgical process itself is proportionally consistent, while planning accuracy is more sensitive to case mix shifts, episodic complexity, and the inherent difficulty of predicting very short and very long procedures. The next sections examine these dynamics in more detail by analysing each duration group separately and comparing the distribution of coefficients of variation across operating rooms.

Surgeries planned under 30 minutes

Short procedures, typically representing high volume and standardized interventions, show moderate relative variability across operating rooms. For this category, coefficients of variation of observed durations range roughly between 0.35 and 0.63, which reflects the proportional impact of small absolute timing fluctuations on short cases. These values are expected for brief procedures, where even limited variation translates into relatively large coefficients once expressed relative to their mean duration. Table 14 summarises these indicators for all rooms with at least eleven surgeries in this duration group.

The distribution across rooms suggests a broadly homogeneous pattern. Units such as LP01, LP02, and LO05 show higher CV values between 0.45 and 0.62, while others including LO02, LO04, LO01, and LO03 show lower variability, with values between 0.35 and 0.36. These differences are small in absolute terms and mainly reflect differences in the mix of short procedures performed in each room rather than substantive disparities in timing consistency.

For the coefficient of variation of planning deviation, values are systematically higher and span a wider range, with most rooms falling between approximately 0.9127 and 1.213. This confirms that relative fluctuations in planning accuracy exceed those associated with the observed durations themselves. The lowest CV deviation values appear in rooms such as LP02, LO01, and LO04, while higher values are found in rooms including LO02, LO05, and LO03.

Overall, short procedures are characterized by consistent execution times and broadly comparable planning accuracy across operating rooms. The relatively high coefficients primarily reflect proportional scaling effects inherent to short interventions rather than substantial operational variability. Subsequent sections examine whether similar patterns hold for procedures of longer planned duration, where absolute variability becomes more dominant and planning precision potentially more challenging.

Table 14: Coefficient of variation per operating room for surgeries planned <30 minutes (only rooms with >10 surgeries shown) (continued below)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev
LP01	20114	14.73	6.893	0.468	4.787	4.664
LP02	18090	15.61	7.157	0.4585	5.157	4.707
LO02	6504	25.09	8.913	0.3552	5.905	5.948
LO04	2714	29.12	10.28	0.3529	7.336	7.069
LO05	1291	22.26	13.96	0.6269	6.697	8.123
LO03	707	29.84	10.52	0.3525	7.395	7.17
LO01	640	30.76	11.04	0.3589	8.483	8.032

CV dev
0.9743
0.9127
1.007
0.9636
1.213
0.9696
0.9469

Surgeries planned 31–60 minutes

Procedures with a planned duration of less than 30 minutes show relatively high proportional variability across operating rooms, reflecting the sensitivity of short procedures to even small absolute fluctuations. For this category, coefficients of variation of observed durations range from about 0.35 to 0.63, as shown in Table ???. The lowest values appear in LO02 (0.3552), LO04 (0.3529), and LO01 (0.3589), while rooms such as LP01 (0.468) and LP02 (0.4585) fall in the mid-range. LO05 shows the highest CV_duration (0.6269), which likely reflects a more heterogeneous short-procedure mix rather than operational inconsistencies.

For the coefficient of variation of planning deviation, values are again higher across all rooms, ranging from about 0.91 to 1.21. As shown in Table ??, the lowest CV_deviation values appear in LP02 (0.9127), LO01 (0.9469), and LO04 (0.9636), while LO03 (0.9669) and LP01 (0.9743) lie near the centre of the distribution. The highest values occur in LO02 (1.007) and LO05 (1.213), indicating that planning deviations for the shortest procedures often match or exceed their mean, which is expected because even a few minutes of difference represent a large proportional deviation.

Taken together, these results show that procedures planned for less than 30 minutes display predictable execution in absolute terms, but inherently high relative variability in both duration and planning deviation. The observed coefficients largely reflect the scaling effects typical of very short procedures rather than meaningful operational differences between rooms.

Table 16: Coefficient of variation per operating room for surgeries planned 31–60 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
LO02	3085	46.58	16.21	0.348	10.08	8.62	0.8553
LO01	2915	44.35	15.1	0.3406	9.718	8.979	0.9239
LO05	2175	41.6	15.68	0.377	9.789	9.248	0.9448

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
LO04	2172	41.38	16.62	0.4017	9.151	8.786	0.9602
LO03	1467	43.8	17.72	0.4045	11.2	10.98	0.9803
LP02	437	27.64	14.72	0.5326	17.72	9.466	0.5341
LP01	64	28.38	12.31	0.434	17.88	9.297	0.5201

Surgeries planned 61–90 minutes

Procedures with a planned duration of 31–60 minutes show moderate relative variability across operating rooms. For this category, coefficients of variation of observed durations cluster between about 0.34 and 0.53, as shown in Table 16. The lowest values appear in LO01 (0.3406) and LO02 (0.348), followed closely by LO04 (0.4017) and LO03 (0.4045). Higher variability is observed in LO05 (0.377) and LP01 (0.434), while LP02 shows the highest CV_duration at 0.5326. These differences likely reflect variation in the underlying case mix rather than structural workflow differences between rooms.

For the coefficient of variation of planning deviation, values are again higher across all rooms, ranging from approximately 0.52 to 0.98. The lowest CV_dev values appear in LP01 (0.5201) and LP02 (0.6341), while rooms such as LO01 (0.9239), LO05 (0.9448), LO03 (0.9803), and LO04 (0.9602) display substantially higher dispersion. LO02 (0.8553) lies in the mid-range. These patterns indicate that relative variation in planning accuracy generally exceeds the variation in the durations themselves, especially in the general-purpose rooms where procedure heterogeneity is greater.

Taken together, these results show that procedures planned for 31–60 minutes display consistent execution times and moderate variation in planning accuracy across operating rooms. The observed coefficients largely follow proportional scaling effects typical of medium-length procedures rather than substantial operational differences.

Table 17: Coefficient of variation per operating room for surgeries planned 61–90 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
LO01	1312	67.88	20.3	0.2991	15.44	12.2	0.7901
LO05	1154	79.28	24.02	0.3029	18.96	14.74	0.7777
LO03	759	84.53	27.85	0.3295	19.81	16.72	0.8441
LO04	639	83.87	29.41	0.3506	19.53	18.92	0.969
LO02	354	59.84	20.3	0.3392	15.75	13.66	0.8672
LP02	92	45.33	15.87	0.3502	27.8	12.36	0.4444

Surgeries planned 91–180 minutes

Surgeries with a planned duration between 91 and 180 minutes include a broad mix of major inpatient operations and complex elective cases. Within this category, the coefficient of variation of observed durations is moderate, ranging mostly between 0.27 and 0.41 across operating rooms, as shown in Table 18. The lowest CV values are observed in LO04 (0.2768) and LO03 (0.2909), indicating a tightly clustered distribution of operative times. Slightly higher values appear in LO05 (0.3322) and LO02 (0.3759), while LO01 shows the highest proportional variability in this group with a CV of 0.4078. These differences remain limited and suggest proportionally consistent execution times despite the broader absolute duration range associated with longer procedures.

The coefficient of variation of planning deviation shows a similarly narrow dispersion. Most rooms fall between 0.76 and 1.01, with the lowest values seen in LO02 (0.7677) and LO04 (0.8273). LO05 (0.8459) and

LO03 (0.8973) occupy the mid-range, while LO01 shows the largest relative dispersion in planning deviation (1.005). Despite these differences, the overall spread remains constrained, indicating stable planning accuracy across rooms even as case mix becomes more heterogeneous.

Across operating rooms, both indicators show restrained variability, implying that once surgeries reach this duration range, timing performance becomes comparatively stable. In contrast to shorter procedures, proportional variation in duration and planning deviation becomes smaller relative to total operative time. These mid- to long-duration operations therefore represent one of the most consistent segments of the surgical program in terms of timing reliability.

Table 18: Coefficient of variation per operating room for surgeries planned 91–180 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
LO03	1251	108.9	31.68	0.2909	22.1	19.83	0.8973
LO05	573	109.1	36.23	0.3322	25.3	21.4	0.8459
LO04	482	105.4	29.17	0.2768	22.67	18.75	0.8273
LO01	326	114.5	46.68	0.4078	26.8	26.94	1.005
LO02	29	105.8	39.78	0.3759	34.79	26.71	0.7677

Surgeries planned over 180 minutes

Procedures with a planned duration exceeding three hours represent the most complex and resource intensive part of the surgical program. Within this group, the coefficient of variation of observed durations remains moderate, ranging from 0.67 to 0.80 across operating rooms, as shown in Table 19. The lowest CV value appears in LO03 (0.6698), while LO05 shows the highest proportional variability at 0.7981. Interpretation remains cautious because both rooms have relatively small case counts, which can amplify variability in these proportional measures.

The coefficient of variation of planning deviation is generally close to or above one. The rooms fall between 0.8391 and 1.027, with LO03 showing the lowest value (0.8391) and LO05 the highest (1.027). These differences reflect the wide absolute spread in deviations that naturally accompanies long and complex procedures rather than systematic inconsistencies in planning accuracy.

Overall, surgeries longer than three hours display stable and proportionally uniform duration patterns across the operating complex. Observed durations show limited relative variability, and planning deviations exhibit consistent proportional spread. The higher coefficients observed in a few rooms likely result from the concentration of rare or high complexity procedures rather than persistent scheduling issues.

Table 19: Coefficient of variation per operating room for surgeries planned > 180 minutes (only rooms with >10 surgeries shown)

OR	n	Mean duration	SD duration	CV duration	Mean dev	SD dev	CV dev
LO05	13	112.3	89.64	0.7981	298.4	306.4	1.027
LO03	12	104.2	69.82	0.6698	365.6	306.7	0.8391

Relationship between variability and mean operative duration

Examining how variability scales with operative duration provides insight into whether dispersion in surgical times increases proportionally with the length of a procedure. By analysing this relationship at the level of procedure types, it becomes possible to distinguish structural scaling effects from differences in planning

performance. The following figures present three complementary perspectives: the standard deviation of operative duration, the coefficient of variation of observed duration, and the coefficient of variation of planning deviation. Together, they describe how both absolute and relative variability evolve as surgeries become longer or more complex.

The first figure shows a strong positive linear relationship between mean operative time and its standard deviation, as illustrated in Figure 9. As the average duration increases, the absolute spread of recorded times expands almost proportionally. This scaling effect is expected, since longer procedures allow for greater absolute variation while maintaining a similar proportional spread. For instance, procedures with a mean duration near one hundred minutes typically show standard deviations around forty to fifty minutes, while operations exceeding three hundred minutes often vary by more than one hundred minutes. This pattern confirms that absolute variability grows with case length without implying greater relative unpredictability.

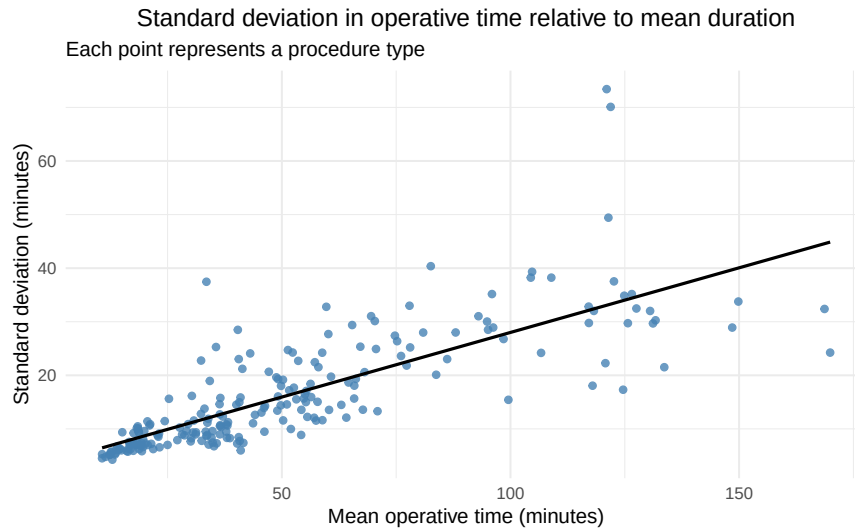


Figure 9: Standard deviation in operative time relative to mean duration

When variability is expressed relative to mean duration, a clear inverse pattern emerges, as illustrated in Figure 10. The coefficient of variation decreases gradually as procedures become longer, indicating that proportional dispersion diminishes with case length. Short procedures often show CV values above 0.4, while most long operations fall below 0.3. This relationship reflects a simple scaling property, where longer procedures display greater absolute variability but proportionally more stability compared with their expected duration.

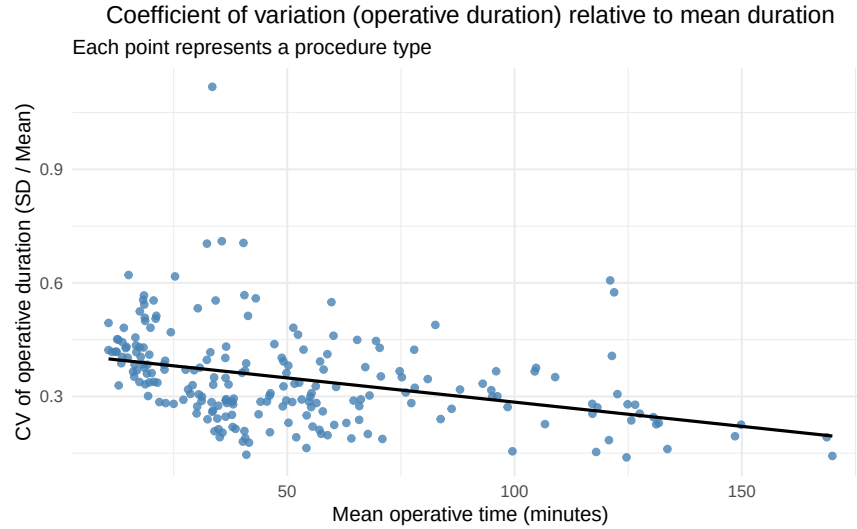


Figure 10: Coefficient of variation (operative duration) relative to mean duration

A similar downward trend is visible when plotting the coefficient of variation of planning deviation against mean duration, as shown in Figure 11. For shorter procedures, the relative spread in planning deviations is wide, with CV values sometimes exceeding 3.0 to 5.0. These high coefficients occur because small absolute deviations represent large proportional differences when the planned duration is short. As procedure length increases, this proportional effect diminishes, and the CV of planning deviation stabilizes around 1.0 or lower. The negative slope therefore reflects the same scaling principle observed for operative duration, rather than a systematic improvement in planning accuracy.

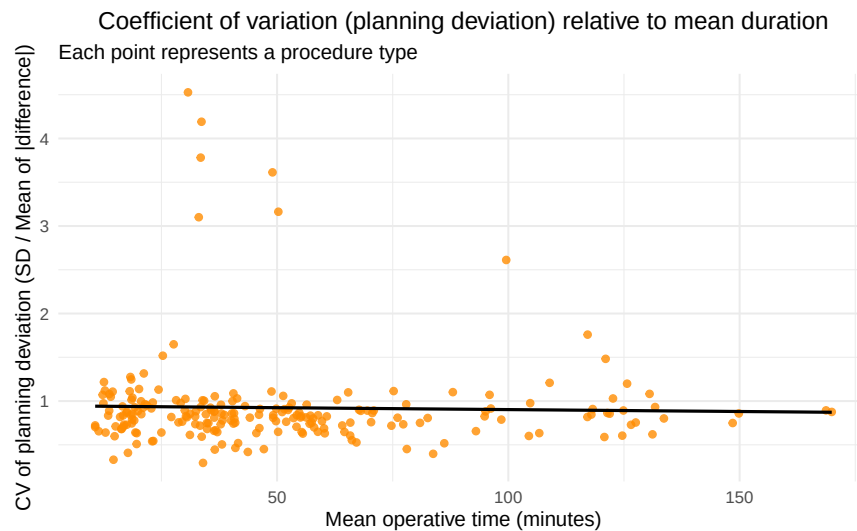


Figure 11: Coefficient of variation (planning deviation) relative to mean duration

Taken together, the three figures show that while absolute variability in surgical time increases with case length, relative variability tends to decrease. This pattern indicates that proportional uncertainty is greatest for short procedures, where small timing differences represent a large share of the total duration, and least for long surgeries, where absolute deviations are diluted across extended operating periods.

How large are start-time delays, and when do they happen?

Start time deviations show a wide spread, with late starts occurring slightly more often than early ones, as summarised in Table 20. About 56.8 percent of all surgeries begin later than scheduled, with an average delay of 32.2 minutes and a median of 21 minutes. Early starts occur in 43.2 percent of cases, with an average lead of 33.5 minutes and a median of 20 minutes. The dispersion is considerable: standard deviations reach 38.8 minutes for late starts and 47.3 minutes for early starts, illustrating substantial variability in when cases begin. A few extreme outliers exceed 1,100 minutes in the late direction, highlighting the presence of rare but very large deviations.

These results confirm that punctuality at the start of the surgical schedule remains a recurrent operational challenge. Even moderate delays early in the day can cascade into later sessions, reducing scheduling flexibility and overall room efficiency.

Table 20: Difference between actual and planned start time (absolute minutes), separated by direction

direction	Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9	Percent
Late start	32.2	21	38.8	32	1	1119	99	186	338.2	56.8%
Early start	33.5	20	47.3	34	0	822	113	225	462	43.2%

The distribution of start-time differences is centered around zero, as shown in Figure 12, with a clear peak at small deviations, indicating that most surgeries start close to their planned time. However, the shape of the histogram is slightly skewed toward positive values, showing that late starts are more frequent than early ones. Most cases begin within approximately half an hour of schedule, but the gradual widening of the distribution toward the right reveals that a smaller subset experiences significant delays.

The presence of a long right-hand tail confirms that occasionally, extreme delays still occur. These outliers have a limited frequency but can affect overall operating room throughput and schedule stability. The symmetrical appearance around the center, combined with a mild skew, suggests that deviations are not random but follow a consistent pattern of minor delay accumulation across the day.

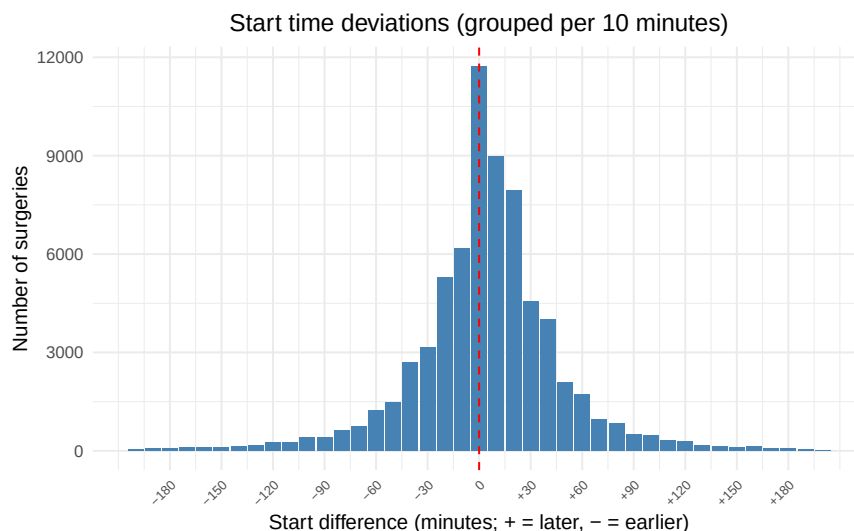


Figure 12: Start time deviations (grouped per 10 minutes)

Average start-time delays show moderate variation across weekdays, as illustrated in Figure 13, ranging from minus 1.8 to 7.4 minutes during regular working days. Monday and Tuesday have mean delays of

7.4 and 6.1 minutes, while Wednesday reaches 6.8 minutes. Thursday and Friday follow with minus 0.7 and minus 1.8 minutes, respectively. These differences indicate that start-time punctuality is relatively consistent throughout the first three days of the week, with fluctuations on Thursday and Friday that are likely to reflect structural scheduling issues.

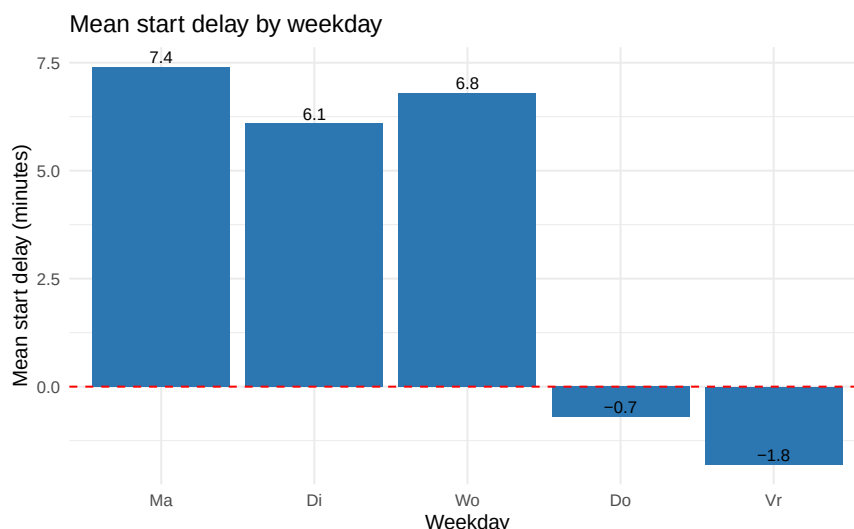


Figure 13: Mean start delay by weekday

Late starts occur more frequently than early starts on every day of the week, as summarized in Table 21. Across regular weekdays, about three fifths of surgeries begin later than planned, with mean delays between 28.6 and 35.4 minutes and median delays around 18 to 23 minutes. Early starts are less common, representing roughly 37.6 to 46.7% of all cases, and typically begin between 32.1 and 40 minutes before the scheduled time. These small variations between weekdays indicate that overall punctuality during the workweek is relatively stable, with no clear day standing out as an outlier.

Table 21: Start-time deviation patterns per weekday (minutes).

Weekday	n	Late starts (%)	Mean delay	Median delay	Early starts (%)	Mean lead	Median lead
Ma	14710	59.7	32.8	21	37.9	-32.1	-19
Di	14902	60.1	35.4	23	37.8	-40	-22
Wo	13742	60	32.1	21	37.6	-33.2	-20
Do	13233	51.7	28.6	18	45.9	-33.8	-22
Vr	12808	51.2	30.7	20	46.7	-37.6	-23

Start-time reliability varies widely between operating rooms, as shown in Table 22. The proportion of late starts ranges from 47.5% to 66%, indicating that punctuality is evenly distributed across the OR complex. Some rooms, such as LO04 and LP01, stand out with high shares of late starts, 66% and 63.5%, respectively. In LO03, the mean delay among late cases reaches 50.5 minutes, driven by a small number of outliers that strongly influence the average.

At the opposite end, rooms such as LO01 and LP02 display a balanced distribution, where early starts occur roughly as often as late ones. These contrasts suggest that certain rooms operate under more stable and predictable timing conditions, while others experience recurring or structural delays. Overall, start-time deviations appear concentrated in a limited set of operating rooms, where occasional long postponements account for much of the overall variability in punctuality.

Table 22: Start-time deviation patterns per operating room (minutes).

ActualOR	n	Late starts (%)	Mean delay	Median delay	Early starts (%)	Mean lead	Median lead
LO04	6009	66	43.2	26	32.1	-52.2	-30
LP01	20189	63.5	24.9	20	34.8	-24.7	-16
LO03	4196	59.6	50.5	31	38	-53.2	-31
LO02	9974	57.1	44.7	29	38.4	-56.3	-33
LO05	5206	56.9	41.3	23	40.7	-45.3	-26
LO01	5202	49.9	33.4	21	47.6	-35.3	-20
LP02	18619	47.5	21.1	16	50.9	-26.2	-20

Which procedures deviate most from plan?

Procedures differ noticeably in how accurately their duration is predicted, as shown in Table 23. The twenty procedure types with the largest absolute deviations display mean differences ranging from about 21 to nearly 39 minutes, indicating that for several procedure types the planned duration can diverge meaningfully from actual performance.

The largest deviations appear in VARICES VSM and LASER – MK, where mean absolute differences exceed 38 minutes. These procedures stand out as the most difficult to estimate accurately, likely because their execution times vary with patient-specific factors or case complexity. Several other procedures with moderate case volumes, such as SCHOUDEER, BREDE EXCISIE MALIGNIE, and EXPLORATIEVE LAPARASCOPIE, show deviations in the range of 26 to 29 minutes, reflecting a mix of predictable structure with occasional outliers.

More frequently performed procedures, including VARICES, VARICES LASER, and PEES HAND (VINGER), show smaller deviations around 21 minutes, suggesting that repetition and routine support more accurate planning. The pattern therefore aligns with an experience-based effect: procedures performed often tend to have tighter planning accuracy, while less frequent or clinically variable procedures show greater dispersion.

Across the listed procedures, the share of longer-than-planned cases exceeds the share of shorter ones for most items. This indicates a systematic tendency toward underestimation, although the magnitude varies between procedure types.

Overall, these results show that planning accuracy at Campus Lanaken depends strongly on procedure type. Common, routine interventions tend to follow their planned durations closely, while specialized or more variable procedures exhibit wider deviations even when the average case length is not particularly long.

Table 23: Top 20 procedures by average absolute duration deviation (Lanaken).

ProcedureName	n	Mean abs dev (min)	Longer (%)	Mean + dev (min)	Shorter (%)	Mean - dev (min)
VARICES VSM	49	38.6	73.5	34.4	26.5	-50.3
LASER - MK						
SCHOUDEER	165	28.7	46.1	32.8	52.7	-25.9
ROTATOR CUFF						
ARTROSCOPISCH						
CHIR.IMPLNT.ACT.IMPL66		28	33.3	12.2	66.7	-36
EXPLORATIEVE	47	26.8	12.8	11.3	87.2	-29
LAPARASCOPIE						

ProcedureName	n	Mean abs dev (min)	Longer (%)	Mean + dev (min)	Shorter (%)	Mean - dev (min)
BREDE EXCISIE MALIGN MELANOOM MET SENT KLIERBIOPS CLAVICULA FRAKTUUR AO KLEIN FRAGMENT	44	26.5	63.6	34.8	34.1	-12.7
Ganglion van Gasser : radiofrequente R HEELKUNDIGE CORRECTIE VAN GYNAECOMASTIE REDUCERENDE MAMMAPLASTIEK	47	26.4	53.2	27.7	44.7	-26.2
LITTEKENCORRECTIE ABDOMENPLASTIE	88	26.1	1.1	4	98.9	-26.3
HAND DUPYTREN DESCEMET'S STRIPPING AUTOMATED ENDOTHELIAL KERTOPLASTY - ENDOTHEELTRANS- PLANTATIE	58	25.8	62.1	29.5	36.2	-20.7
epidurale stimulatie:diagnostisch lumbaal enkele elektrode Ganglion	73	24.6	46.6	29	50.7	-21.8
sfenopalatinum : Radiofrequente R PEES HAND (VINGER) VOET HECHTING	65	24.1	53.8	31.2	41.5	-17.5
VARICES VARICES LASER EXC MALIGN HUIDLETSEL MT RECONST HUIDGR HFD/GEZIC	44	23.4	59.1	25.7	38.6	-21.2
epidurale stimulatie: batterij implantatie	75	22.9	33.3	16.4	64	-27.2
	34	22.3	47.1	27.8	50	-18.4
	41	22.1	63.4	26.1	36.6	-15.2
	78	22	0	NA	98.7	-22.3
	46	21.7	26.1	26.6	71.7	-20.5
	689	21.3	52.5	25.2	46	-17.5
	1351	21.3	49.3	24.8	49.4	-18.4
	45	21.2	48.9	23.8	51.1	-18.7
	54	21.1	64.8	25	33.3	-14.9

The scatter plot in Figure 14 shows an inverse relationship between procedure volume and average deviation. Procedures that occur more frequently tend to have lower mean absolute deviations, while rare procedures show much greater variability in duration. This pattern is visible in the downward trend of the points as procedure volume increases.

High-volume procedures, which appear toward the right of the chart, cluster at lower deviation values, indicating more stable and predictable planning performance. In contrast, low-volume procedures on the left display a much wider range of deviations that can exceed 25 minutes. The relationship suggests that

familiarity and repetition contribute to more accurate duration estimates, whereas infrequent procedures are harder to predict due to limited historical data. Overall, the figure highlights that case volume is an important factor in explaining the variability of planning accuracy across procedure types.

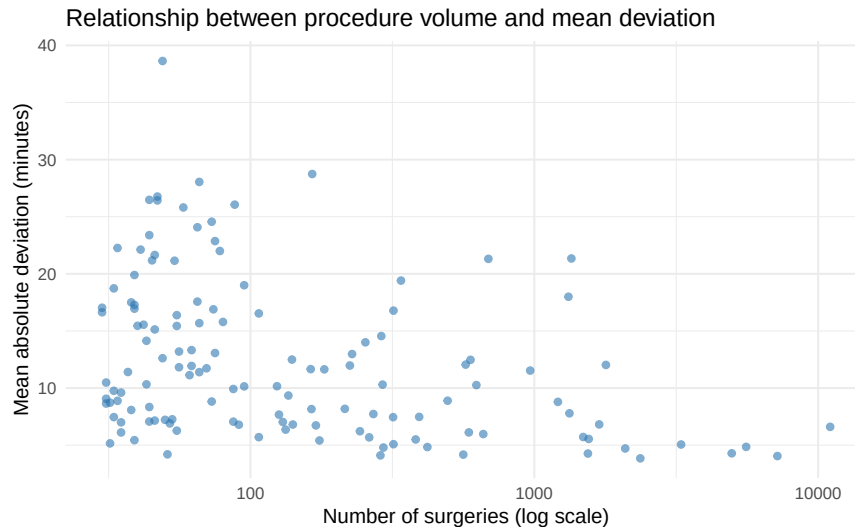


Figure 14: Relationship between procedure volume and mean deviation

The distribution of duration deviations across surgeons reveals notable variation in planning accuracy, as shown in Table 24. Among those with at least fifty recorded procedures, the average absolute deviation ranges from 15.7 to 25 minutes, while the average relative deviation spans 20.2% to 28.5%. A few surgeons show comparatively lower mean deviations below 19 minutes, whereas most fall between 19 and 24 minutes, indicating that planning performance differs across individuals but remains within a moderate range for the majority.

No clear relationship appears between the number of recorded cases and the size of the deviation, suggesting that factors beyond case volume influence the observed variation. Overall, the results show that duration deviations are not uniformly distributed across surgeons, emphasizing that planning accuracy can vary meaningfully even within a shared operational environment.

Table 24: Top 15 surgeons by average absolute and relative duration deviation (pseudonymized).

Surgeon	n	Mean absolute dev (min)	Mean relative dev (%)
Surgeon 1	353	25.0	25.9
Surgeon 2	223	24.5	25.1
Surgeon 3	597	24.2	25.4
Surgeon 4	728	23.9	24.5
Surgeon 5	151	23.3	26.8
Surgeon 6	361	21.8	28.5
Surgeon 7	73	21.6	28.0
Surgeon 8	202	20.5	21.8
Surgeon 9	603	19.0	23.2
Surgeon 10	128	18.7	22.9
Surgeon 11	270	18.6	20.9
Surgeon 12	376	18.4	20.2
Surgeon 13	361	16.5	24.2

Surgeon	n	Mean absolute dev (min)	Mean relative dev (%)
Surgeon 14	132	16.4	21.6
Surgeon 15	297	15.7	25.3

Do surgeries stay within regular working hours?

How often do cases extend beyond their assigned time window?

Only a very small share of surgeries at Lanaken extends beyond the scheduled shift end, as shown in Table 25. Out of nearly seventy thousand procedures, 0.5 percent continue into after-hours, indicating that overtime is an exceptional rather than routine event in this campus. When it does occur, overtime remains limited: the mean extension is 15.7 minutes, and the median is 10 minutes, showing that most overruns are brief and operationally contained. A smaller subset of cases lasts longer, with the 95th percentile reaching 48.5 minutes, illustrating that occasional outliers contribute disproportionately to the total overtime minutes.

These results highlight a scheduling system that operates with very high temporal adherence. The extremely low percentage of after-hours cases suggests that the daily program rarely spills over into the next shift, and when it does, the additional time required is usually modest. The distribution nevertheless shows that a few atypical procedures or unexpected delays can still push individual cases into the after-hours window, but such events remain rare within the overall workflow at Lanaken.

Table 25: Overall share and duration of after-hours surgeries (Lanaken).

Total	Surgeries with overtime	Percentage	Mean Overtime	Median Overtime	P95 Overtime
69395	371	0.5	15.7	10	48.5

The distribution of overtime duration, shown in Figure 15, is strongly right-skewed. Most extended cases end within the first 30 minutes beyond the scheduled shift, and the highest frequency occurs between 0 and 20 minutes of overtime. Beyond that, the number of cases declines steadily, with only a small fraction exceeding 60 minutes. A few extreme cases surpass 100 minutes, but these are rare.

This pattern confirms that while overtime is a recurring operational feature, it is typically limited in duration. The long right tail illustrates that a small group of prolonged procedures accounts for a disproportionate share of total overtime minutes. These outliers can heavily affect staff workload and room turnover late in the day, but they do not represent the typical case. Overall, overtime at Lanaken mainly consists of brief extensions of planned activity rather than frequent, sustained night work.

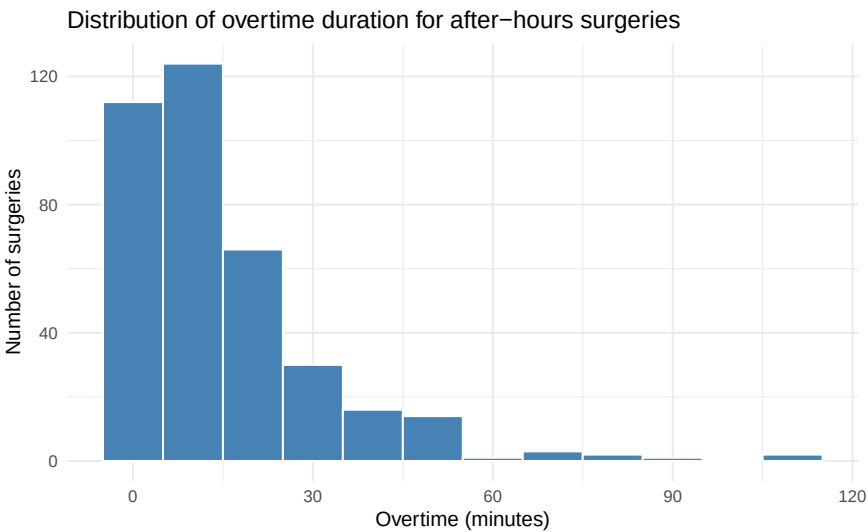


Figure 15: Distribution of overtime duration for after-hours surgeries

The share of surgeries extending beyond scheduled shift hours remains steady throughout the regular work-week, as shown in Figure 16, fluctuating between 0.3 and 0.8 percent from Monday to Friday. This indicates that weekday operations are largely contained within their planned time frames, with only a small proportion of cases requiring additional time beyond shift boundaries.

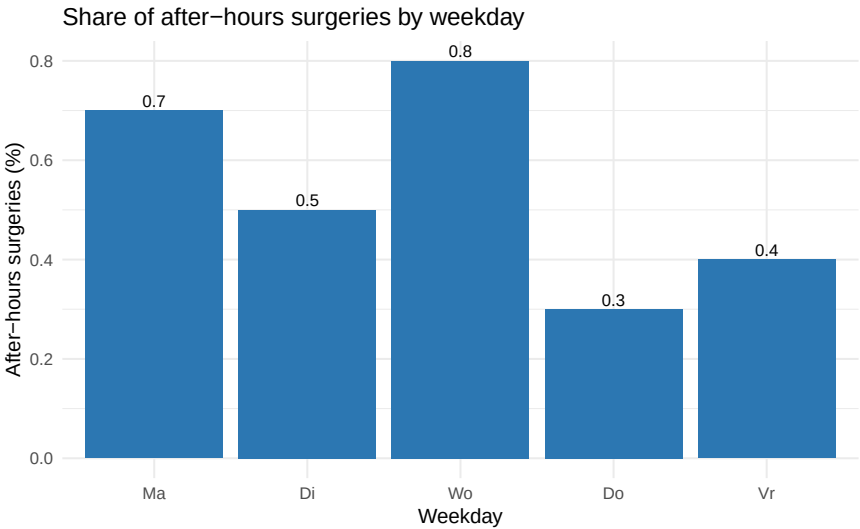


Figure 16: Share of after-hours surgeries by weekday

The proportion of surgeries extending beyond scheduled shift hours remains extremely low and shows no structural change over time, as illustrated in Figure 17. In 2022 and 2023, about 0.5 percent of surgeries at Lanaken exceeded their scheduled shift. This share increases slightly to 0.6 percent in 2024 before returning to 0.5 percent in the partial dataset for 2025.

These values represent the percentage of all procedures performed in each year that continued past the official end of their assigned shift. The near-identical year-to-year percentages demonstrate that after-hours activity at Lanaken is highly stable, with only minimal variation across the observed period.

Because overtime is so rare, these small fluctuations are unlikely to signal operational changes. Instead, they probably reflect routine variability in low-frequency events rather than shifts in throughput, planning accuracy, or case mix. Overall, after-hours work at Lanaken is a consistently marginal component of surgical activity, indicating very strong adherence to scheduled shift boundaries and efficient daily workflow management.

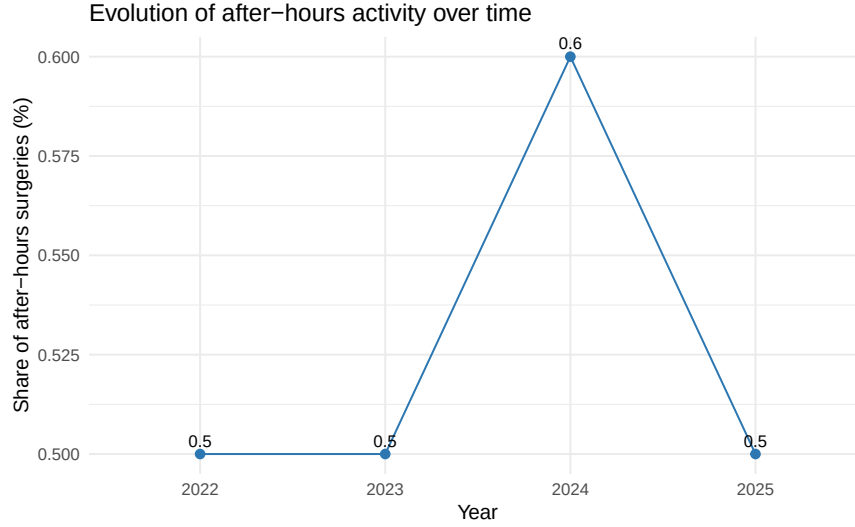


Figure 17: Evolution of after-hours activity over time

The proportion of surgeries extending beyond scheduled shift hours differs markedly across operating rooms, as summarized in Table 26. Most rooms show overtime rates between 0.6 and 1.2 percent, while one stands out with slightly higher shares. LO03 records the highest proportion of overtime cases at 1.7 percent, with a mean duration of 18.9 minutes and a 95th percentile of 51.8 minutes, indicating frequent and occasionally prolonged extensions. Such high figures likely reflect the type of procedures performed in this room, which are often complex and less predictable in length.

Several other rooms, including LO05, LO01, and LO04, show overtime in roughly 1.1 to 1.2 percent of cases, with mean durations between 15.2 and 18 minutes. These values suggest that while most surgeries in these rooms finish within their allocated shift, a small share still extends well into the next operational block.

Overall, the data indicate that overtime at Lanaken is concentrated in a limited number of high-volume or complex operating rooms. While many rooms complete their scheduled activity within the planned shift most of the time, a small subset still experiences overtime in 0.1 to 1.7 percent of surgeries, showing that after-shift extensions remain a recurring part of daily operations rather than isolated exceptions.

Table 26: Operating rooms by after-hours activity (Lanaken).

ActualOR	Cases	Percentage of overtime			
		surgeries	Mean Overtime	Median Overtime	P95
LO03	4196	1.7	18.9	14	51.8
LO05	5206	1.2	18	15	46
LO01	5202	1.1	20.7	15	63.8
LO04	6009	1.1	15.2	9	37.6
LO02	9974	0.6	12	9	38.7
LP01	20189	0.1	9.1	8	19.8
LP02	18619	0.1	7.7	5	19.6

When does overtime occur, and how heavy is it?

The distribution of overtime surgeries, shown in Figure 18, displays a clear concentration of cases ending shortly after 16:30. This peak primarily represents procedures that started during the regular day shift and

finished later than planned. The number of surgeries declines rapidly as the evening progresses, indicating that most cases extending beyond their scheduled shift end within the first one to two hours afterward.

After 22:00, only a smaller number of surgeries continue to appear, suggesting that operations ending during the late evening or night are relatively infrequent. Very few procedures extend past midnight, and only isolated cases finish after 07:00.

Overall, the pattern demonstrates that overtime activity is dominated by operations that conclude soon after the end of the regular working day, while surgeries finishing later in the night represent a much smaller portion of total activity.

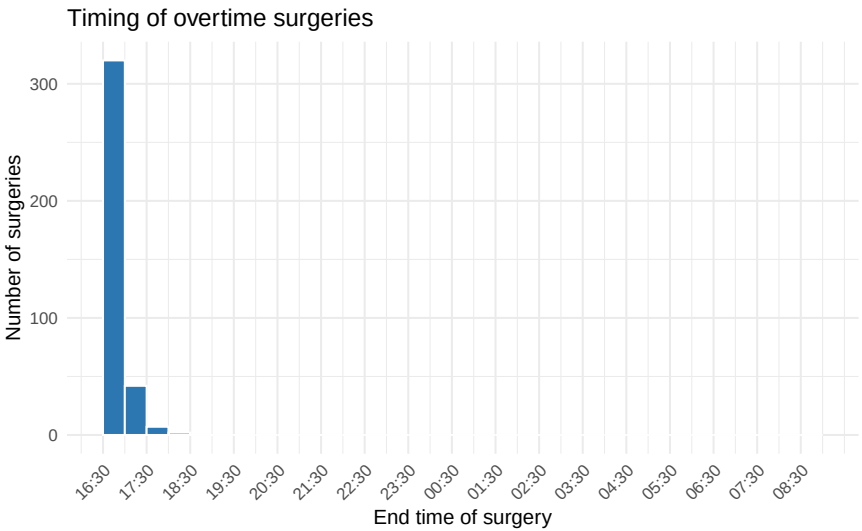


Figure 18: Timing of overtime surgeries

Average overtime durations are highly consistent across weekdays, as illustrated in Figure 19, with means ranging from 13.1 to 18.9 minutes from Monday to Friday. The median per weekday is clearly lower, between 7.5 and 12 minutes, which indicates a right-skewed distribution where a small number of long cases raises the mean.

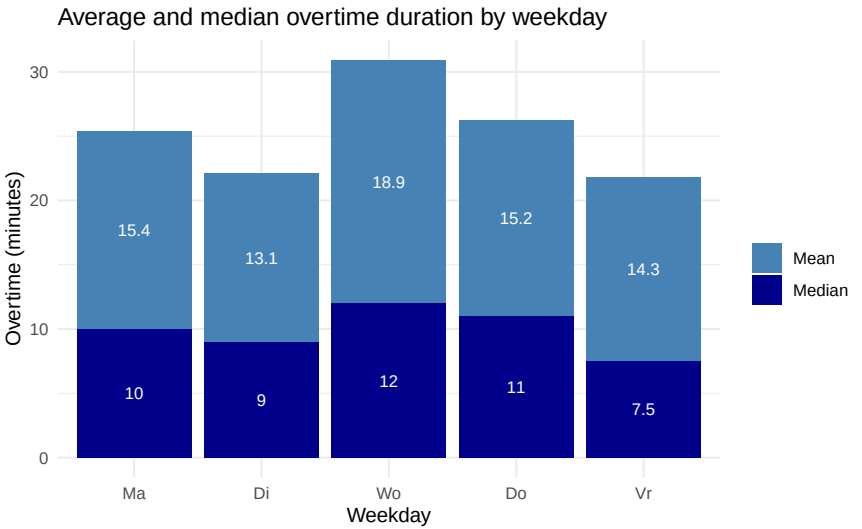


Figure 19: Average and median overtime duration by weekday

Average and median overtime durations per surgery show a largely stable pattern over the observed years, as illustrated in Figure 20. The mean overtime remains close to 14–15 minutes, ranging from 14.1 minutes in 2022 to a temporary rise of 19.2 minutes in 2024, before returning to 14.4 minutes in 2025. The median follows a similar structure at lower levels, fluctuating between 9 and 13.5 minutes.

The consistent gap between the mean and median indicates a right-skewed distribution, where most overtime periods are short but a small number of longer extensions pull the average upward. In practical terms, the typical overtime duration (median) lies around 10 minutes, while the mean reflects the influence of occasional longer cases.

The visible increase in 2024 represents a short-term deviation rather than a structural shift. Across the full period, the overall level of overtime duration remains remarkably stable, with only modest year-to-year variation. As with other campuses, the persistence of a mean exceeding the median underscores that a small minority of prolonged cases accounts for a disproportionate share of total overtime minutes, even in an environment where overtime is rare.

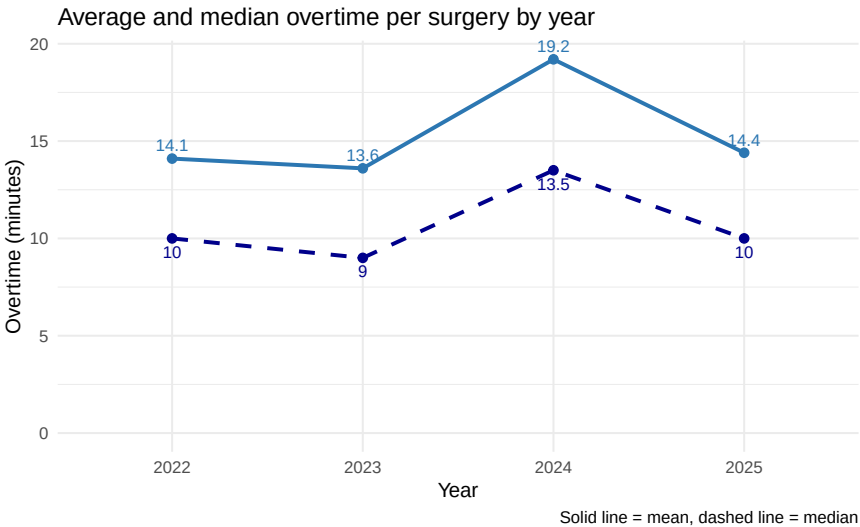


Figure 20: Average and median overtime per surgery by year

The distribution of overtime differs noticeably between shift types, as shown in Figure 21. The day shift (08:00–16:30) displays a wide spread, with most cases extending only slightly beyond the scheduled end but a small number continuing for several hours. This pattern explains why the day shift accounts for most overtime minutes overall.

Overall, the distributions illustrate that overtime is most variable during the day, when activity levels are highest, and more contained during the later shifts. The pronounced right tails in each group confirm that a limited number of long-running cases account for a substantial share of total overtime minutes.

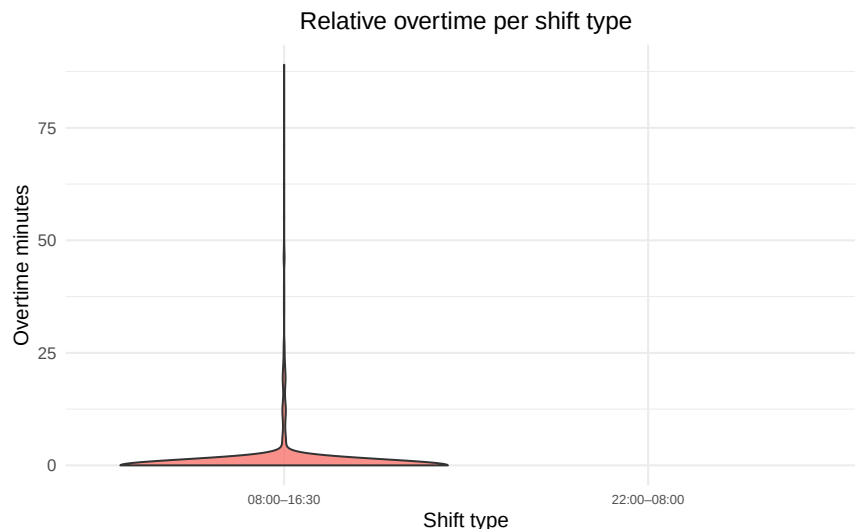


Figure 21: Relative overtime per shift type

How does Lanaken compare with Genk, Cathlab, and Maaseik?

The comparison between campuses reveals pronounced differences in after-hours surgical activity. As shown in Table 27, Genk performs the largest share of overtime surgeries, with 8.4 percent of all cases extending beyond 16:30. Cathlab follows with 5.8 percent, while Maaseik records a more moderate 2.8 percent. Lanaken, by contrast, shows minimal after-hours activity at 0.5 percent, reflecting its lower surgical volume and the limited number of complex or time-intensive cases performed there.

Overtime duration metrics further underline these contrasts. Genk shows the highest average and upper-percentile values, with an average overtime of 59 minutes and a ninety-fifth percentile of 193.8 minutes, indicating that prolonged extensions are not uncommon at the main campus. Maaseik and Cathlab display intermediate levels, while Lanaken again shows the shortest durations, with an average overtime of 15.7 minutes and a ninety-fifth percentile of 48.5 minutes.

Taken together, these figures show that after-hours surgical activity is heavily concentrated at the larger campuses, particularly Genk, while Lanaken contributes only marginally to extended operating room use within the ZOL network.

Table 27: Comparison of after-hours activity between campuses.

Campus	Total	AfterHour surgeries	Share of overtime surgeries	Average Overtime	Median Overtime	P95 Overtime
Genk	96044	8024	8.4	59	38	193.8
Cathlab	9282	539	5.8	41.8	32	108.3
Maaseik	53902	1494	2.8	42	29	129
Lanaken	69395	371	0.5	15.7	10	48.5

Are surgeries performed in their assigned operating rooms?

How frequently are surgeries relocated to a different room?

Room reallocations are relatively rare at Lanaken, as shown in Table 28. Out of 69395 recorded surgeries, around 7.5 percent were performed in a different operating room than originally planned. This low percentage indicates that room assignments are followed very consistently, with only a small number of last-minute adjustments required during daily operations.

Although infrequent, these relocations are operationally important because they typically involve reactive scheduling decisions and resource coordination. Even minor disruptions can have downstream effects on room preparation, staff planning, and turnover time. Tracking their frequency provides useful insight into the stability of the planning process and helps identify whether reassignments occur sporadically or tend to cluster in specific rooms or days.

Table 28: Overall frequency of surgeries relocated to a different room (Lanaken).

Total	Swapped (absolute)	Swapped (%)
69395	5192	7.5

The share of relocated surgeries remains low across all weekdays, as shown in Figure 22, generally fluctuating between 5.9 and 10.3 percent. This stability indicates that the room assignment process is robust during regular working days, with only minimal need for unplanned reallocations. The lowest rate is observed on Wednesday, while Monday show higher values, of 10.3 percent.

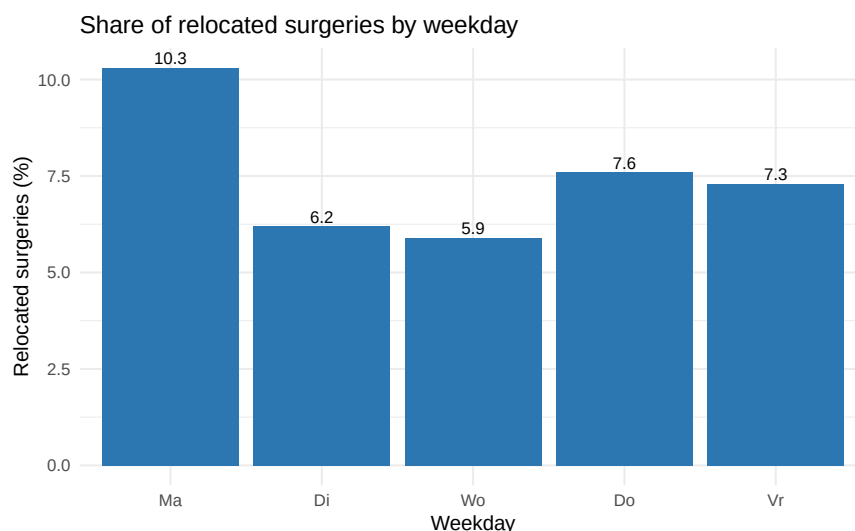


Figure 22: Share of relocated surgeries by weekday

The monthly percentage of surgeries that are relocated to a different operating room fluctuates between 5 and 15.5 percent, as shown in Figure 23. Despite short-term variation, no clear upward or downward trend is visible over the observed period. Occasional peaks above 10 percent indicate that some months experience more frequent reallocations, but these appear as structural changes rather than isolated events.

The relatively low and stable baseline confirms that room swaps are infrequent and mainly driven by temporary scheduling pressures or logistical constraints. The absence of a consistent trend over time suggests that the overall stability of room assignments at Lanaken has remained steady throughout the study period.

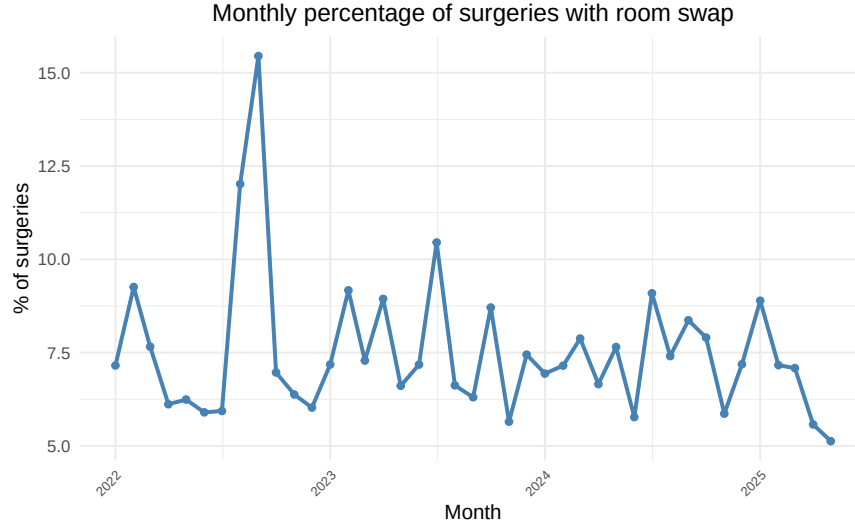


Figure 23: Monthly percentage of surgeries with room swap

Are specific rooms affected more than others?

The analysis shows that only a limited number of operating rooms are regularly involved in relocations, as summarized in Table 29. A small subset of rooms accounts for nearly all movements, while most others experience swaps only sporadically. The most prominent interaction occurs between LP02 and LP01, which exchange cases extensively: every surgery moved out of LP02 is reassigned to LP01, and vice versa. This symmetry suggests that the two rooms function as a tightly coupled pair with interchangeable capacity, absorbing short-term adjustments when local scheduling pressure arises.

Within the main operating block, LO03 and LO04 display a similar pattern, frequently receiving each other's relocated cases. LO03 also sends a smaller share of cases to LO01 and LO02, indicating broader compatibility with several rooms. LO05 participates in relocations as well, though at a lower frequency, generally receiving overflow from LO03 and LO04 rather than acting as a major source of swaps.

These consistent patterns show that reallocations follow clear operational structures rather than occurring randomly. Rooms with similar procedural profiles or shared staffing arrangements naturally form relocation clusters. Movements therefore reflect adaptive use of available capacity to maintain workflow continuity, rather than instability in the planning process.

Table 29: Top 10 planned ORs with the highest number of room swaps, showing the five most frequent new OR destinations (Lanaken).

PlannedOR	TotalMoved	Top1	Top2	Top3	Top4
LP02	2389	LP01 (2389, 100%)	NA	NA	NA
LP01	1140	LP02 (1140, 100%)	NA	NA	NA
LO03	588	LO04 (248, 42.2%)	LO05 (238, 40.5%)	LO01 (93, 15.8%)	LO02 (9, 1.5%)
LO05	431	LO03 (242, 56.1%)	LO04 (121, 28.1%)	LO01 (65, 15.1%)	LO02 (3, 0.7%)
LO04	358	LO03 (216, 60.3%)	LO05 (96, 26.8%)	LO02 (37, 10.3%)	LO01 (9, 2.5%)

PlannedOR	TotalMoved	Top1	Top2	Top3	Top4
LO01	184	LO03 (94, 51.1%)	LO05 (70, 38%)	LO04 (13, 7.1%)	LO02 (7, 3.8%)
LO02	100	LO04 (87, 87%)	LO03 (9, 9%)	LO01 (2, 2%)	LO05 (2, 2%)
LOR0	2	LO01 (1, 50%)	LO05 (1, 50%)	NA	NA

A small number of operating rooms receive the majority of relocated surgeries, as shown in Table 30 and Figure ?? . LP01 and LP02 stand out clearly: LP01 alone accounts for 3.44 percent of all relocated cases, and LP02 adds another 1.64 percent. These two rooms function as the primary destinations when schedule adjustments require shifting procedures, reflecting their flexible capacity and broad procedural compatibility.

Several rooms in the main surgical block also appear regularly as secondary destinations. LO03, LO04, and LO05 together receive a nontrivial share of relocated cases, though far below the volumes seen in LP01 and LP02. At the lower end of the distribution, rooms such as LO01 and LO02 receive only a small number of relocated procedures.

Both the table and figure reveal a highly concentrated pattern: a small subset of rooms absorbs most reallocations, while the remainder sees minimal or occasional inflow. This concentration suggests that relocation decisions are not random but guided by practical constraints such as equipment availability, staffing configuration, and the capacity of specific rooms to act as operational buffers.

Table 30: Top 15 actual operating rooms most frequently used as destinations after relocation (Lanaken).

ActualOR	ReceivedCases	SharePct
LP01	2389	3.44
LP02	1140	1.64
LO03	561	0.81
LO04	469	0.68
LO05	407	0.59
LO01	170	0.24
LO02	56	0.08

Procedures with the highest likelihood of being relocated mainly consist of short or outpatient interventions, as shown in Table 31. The top of the list is dominated by minor, standardised procedures such as CARPAAL TUNNEL, Echo nervus cutaneus femoralis lateralis PRF, and Echo perifere zenuw PRF, which show relocation rates ranging from about 17 to 47 percent. These values are several times higher than the overall relocation rate at Lanaken of 7.5 percent, indicating that a small subset of procedures accounts for a disproportionate share of all room swaps.

These high-swap procedures share an important operational characteristic: they are short, predictable, and compatible with multiple procedure rooms. This makes them particularly suitable for reassignment when minor scheduling conflicts or capacity pressures arise. In contrast, longer or more specialised procedures rarely appear among relocated cases, which reflects their greater dependence on specific room configurations or equipment.

Overall, the table shows that relocations at Lanaken are concentrated in a narrow group of routine, highly transferable procedures. This pattern suggests that room swaps do not reflect instability in planning but rather the deliberate use of flexible procedure types to absorb day-to-day scheduling adjustments.

Table 31: Top 15 procedure types with the highest share of relocated surgeries (Lanaken).

ProcedureName	n	SwapPct
CARPAAL TUNNEL	1211	47.07
Echo nervus cutaneus femoralis lateralis PRF	52	42.31
Echo perifere zenuw PRF	73	38.36
Echo bursa trochanterica	170	29.41
VARICES	689	25.98
Endoscopische carpaal tunnel	124	25.81
NL RF facet lumbaal	39	23.08
Echo nervus genicularis RF (WOMAC)	140	22.86
VARICES LASER	1351	22.65
ZENUW CARPAAL TUNNEL	495	21.21
Ganglion sfenopalatinum : Radiofrequente R	78	20.51
HAND CYSTE	62	19.35
Echo buikwand/ACNES PRF	32	18.75
VINGER TRIGGERVINGER	228	17.62
Echo nervus ilio-inguinalis PRF	35	17.14

What is the operational impact of these reallocations?

Relocated surgeries differ from non-relocated ones in several aspects of timing and scheduling performance, as summarized in Table 32. For start times, relocated cases show a higher average delay among late starts (39.4 minutes versus 31.7 minutes) and a larger lead among early starts (49.1 minutes versus 33.8 minutes). This suggests that these surgeries are more often scheduled under atypical or time-constrained circumstances, which can result in greater variability in when they actually begin. The median values, however, remain closer together (26 versus 20 minutes for late starts, 36 versus 20 for early starts), indicating that while some relocated surgeries start much later or earlier than expected, most are only moderately affected.

For procedure duration, both relocated and non-relocated surgeries show comparable deviations from their planned times. Average differences are around 9 to 10 minutes for longer cases and 8 to 9 minutes for shorter ones, suggesting that relocation itself does not substantially influence how long a surgery takes once it begins.

Overtime remains limited in both groups, with median values at zero. Mean overtime is identical between relocated and non-relocated cases (0.1), which indicates that relocations rarely lead to extended working hours.

Overall, the results indicate that relocations are associated with greater variability in start timing but not with systematically longer or shorter surgeries. Their operational impact therefore appears to be primarily related to scheduling and coordination rather than to the procedures themselves.

Table 32: Comparison of start and duration deviations (early/late, shorter/longer) and overtime between relocated and non-relocated surgeries (Lanaken).

Type	Metric	Mean	Median
Relocated	Late start	39.4	26
Not relocated	Late start	31.7	21
	—		
Relocated	Early start	49.1	36
Not relocated	Early start	33.8	20
	—		

Type	Metric	Mean	Median
Relocated	Longer duration	9.9	6
Not relocated	Longer duration	8.9	5
	—		
Relocated	Shorter duration	8.7	5
Not relocated	Shorter duration	7.8	5
	—		
Relocated	Overtime	0.1	0
Not relocated	Overtime	0.1	0
	—		

Overtime is slightly more common among relocated surgeries than among those performed in their originally assigned room, as shown in Table 33. About 0.5 percent of non-relocated surgeries extend beyond 16:30, compared to 0.4 percent of relocated ones. This small difference likely reflects that some relocated cases are scheduled later in the day or in response to last-minute adjustments, which can occasionally lead to minor overruns.

The similar percentages indicate that room reassignments are not a major driver of evening work. Although relocations require coordination, they do not substantially contribute to overtime or after-hours workload at the campus level.

Table 33: Share of after-hours surgeries by relocation status (Lanaken).

SwapGroup	AfterHours	Percentage of surgeries with afterhours
Not relocated	348	0.5
Relocated	23	0.4

The distribution of overtime duration for relocated surgeries, shown in Figure 24, is highly concentrated near 10 to 20, indicating that many of these cases finish slightly after regular working hours or shortly afterward. A steep drop is visible beyond the first 20 minutes of overtime, and only a few relocated surgeries extend beyond 40 minutes. The long but sparse tail confirms that extreme cases are rare.

This pattern supports the earlier finding that relocations mostly involve short procedures that can be completed quickly once a new room becomes available. The limited overtime associated with these cases suggests that room reassignments are generally managed efficiently and do not significantly contribute to prolonged evening work. In operational terms, relocated surgeries appear to represent flexible workload adjustments rather than sources of additional delay.

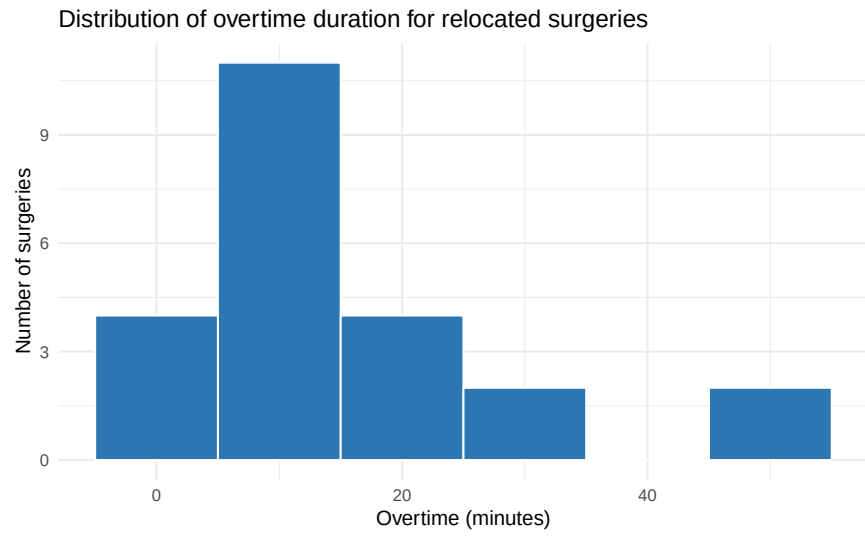


Figure 24: Distribution of overtime duration for relocated surgeries

How do urgent cases disrupt elective planning?

When and where do urgent surgeries occur?

The majority of surgeries at Lanaken are elective, representing about 99.6 percent of all recorded cases, as shown in Table 34. Non-elective or urgent surgeries make up the remaining 0.4 percent of the total. This indicates that most activity follows a planned schedule, only a tiny portion of cases still arises from unplanned or time-sensitive demands.

The relative size of the non-elective segment highlights its operational significance. Even though urgent cases form a minority, their unpredictable timing can affect the stability of daily planning and room allocation. Understanding when and where these non-elective procedures occur provides an important basis for evaluating how they interact with elective activity and whether they contribute to schedule deviations or after-hours work.

Table 34: Distribution of surgeries by urgency classification (Lanaken).

UrgencyType	Total	SharePct
electief	69109	99.6
niet-electief	286	0.4

The start times of non-elective surgeries, shown in Figure 25, display a clear concentration during daytime hours, with a pronounced peak between 13:00 and 15:00. This pattern indicates that the majority of urgent procedures are performed within or near regular operating hours, often overlapping with scheduled elective activity.

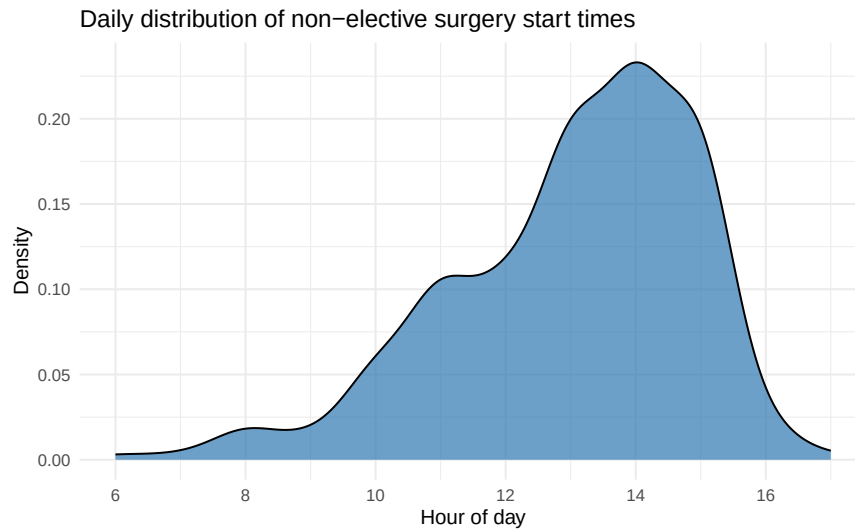


Figure 25: Daily distribution of non-elective surgery start times

Across all weekdays, non-elective surgeries follow a consistent hourly pattern, as illustrated in Figure 26. Most activity occurs between late morning and early evening, with peaks around 13:00 to 15:00 on nearly every day. This confirms that urgent procedures are frequently performed during regular daytime hours. The overlap with elective schedules suggests that urgent cases often need to be accommodated within existing capacity rather than outside normal shifts.

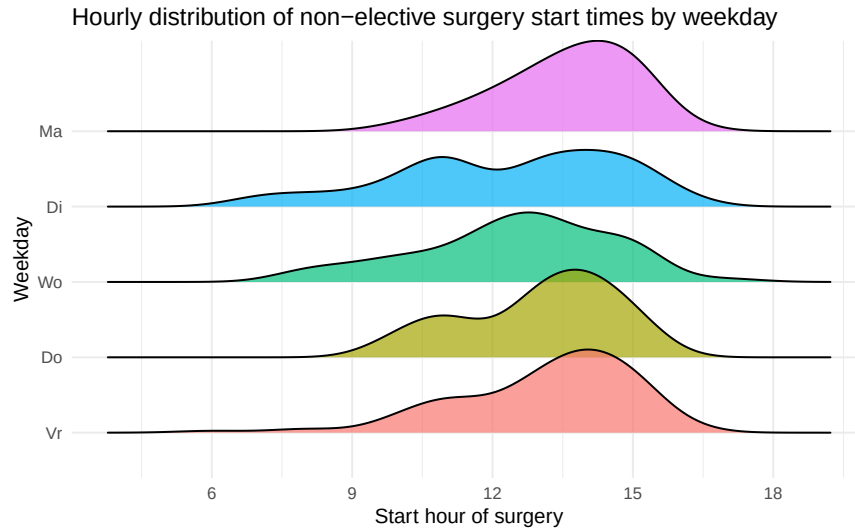


Figure 26: Hourly distribution of non-elective surgery start times by weekday

Elective surgeries dominate the activity, as shown in Table 35, accounting for roughly 99 percent of all cases from Monday to Friday. The share of non-elective procedures remains stable during the workweek, between 0.2 and 0.8 percent, indicating a consistent and predictable pattern of urgent demand. This stability suggests that urgent cases are routinely managed alongside scheduled operations without major fluctuations across days.

Table 35: Weekday distribution of elective vs. non-elective surgeries (Lanaken).

Weekday	Elective (n)	Non-elective (n)	Elective (%)	Non-elective (%)
Ma	14635	75	99.5	0.5
Di	14878	24	99.8	0.2
Wo	13683	59	99.6	0.4
Do	13209	24	99.8	0.2
Vr	12704	104	99.2	0.8

Do urgent cases drive overtime?

Urgent cases contribute to overtime work but are not its primary driver, as shown in Table 36. While the share of non-elective surgeries extending beyond the shift is higher (3.8 percent) than for elective ones (0.5 percent), the overall difference remains moderate. This indicates that both types of activity continue after the shift, although elective cases represent the majority in absolute numbers due to their much larger total volume.

Average overtime per case is slightly higher for non-elective surgeries, at 0.4 minutes compared with 0.1 minutes for elective ones. The 95th percentile follows a different pattern, reaching 0 minutes for both urgent cases and for elective ones. This suggests that urgent procedures, much less frequent, barely require longer extensions when they do occur.

Overall, the results imply that unplanned demand does contribute to extended operating hours, but most work still stems from elective cases that overrun their planned schedules rather than from late-arriving emergencies.

Table 36: After-hours frequency and overtime duration by urgency type (Lanaken).

Urgency type	Total (n)	After-hours (n)	After-hours (%)	Mean overtime (min)	95th percentile (min)
Elective	69109	360	0.5	0.1	0
NonElective	286	11	3.8	0.4	0

Do urgent cases interrupt elective planning?

The results in Table 37 show that the frequency of room swaps among elective surgeries is virtually identical regardless of whether an overlap with a non-elective case occurred. For surgeries not affected by an overlap, 7.5 percent involved a room swap, compared to the 3.5 percent among overlapped cases.

This consistency indicates that while urgent procedures occasionally coincide with scheduled elective activity, they do not systematically trigger room reallocations. Instead, such overlaps appear to be managed through local coordination or minor timing adjustments rather than by physically moving elective cases to different operating rooms. Overall, urgent interventions create some scheduling pressure but do not significantly disrupt room allocation patterns within the elective program.

Table 37: Room swap frequency among elective cases: overlapped vs. not overlapped.

Overlapped	N	RoomSwaps	ShareSwapped
No	68652	5169	7.5
Yes	455	16	3.5

As shown in Table 38, overlaps between urgent and elective surgeries occurred on 143 out of 853 observed days, corresponding to 16.8 percent of all days. This moderate frequency indicates that schedule interference is not an exceptional event but a regular characteristic of daily activity. While individual overlaps may vary in magnitude and operational impact, their near-daily occurrence highlights the continuous need for flexibility in resource allocation and real-time coordination between planned and urgent care.

Table 38: Days with at least one urgent–elective overlap in the same OR.

OverlapDays	TotalDays	SharePct
143	853	16.8

The proportion of elective surgeries affected by urgent overlaps fluctuates modestly over time, as shown in Figure 27. Monthly values generally range between 0.4 and 0.8 percent. A gradual upward trend is visible throughout the end of 2023 and 2024. Despite these month-to-month variations, the overall level remains relatively stable, suggesting that the frequency of urgent cases interfering with scheduled activity is a persistent feature of daily operations rather than a short-term anomaly.

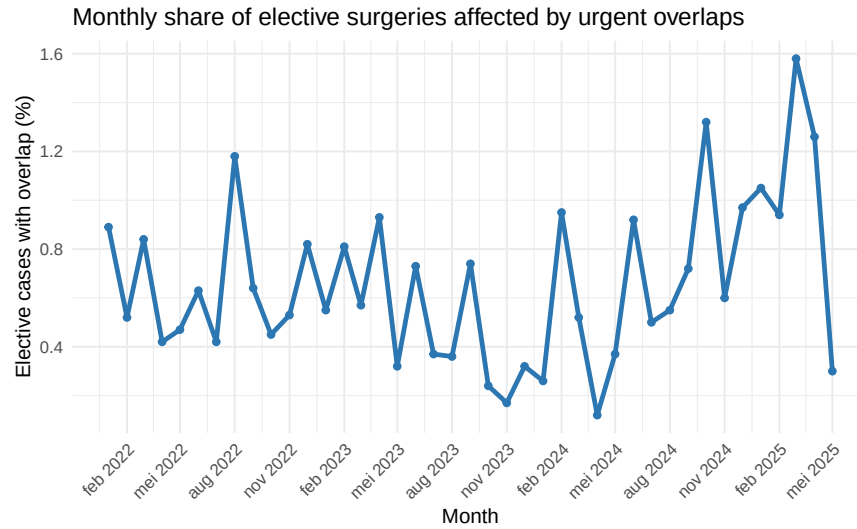


Figure 27: Monthly share of elective surgeries affected by urgent overlaps

As shown in Figure 28, overlaps between urgent and elective surgeries occur during daytime hours. The frequency increases sharply after 12:00, peaks between 13:00 and 15:00, and then declines sharply toward 16:00. The clustering of overlaps in the middle of the day suggests that urgent admissions frequently coincide with the busiest elective periods, intensifying competition for operating room capacity during standard working hours.

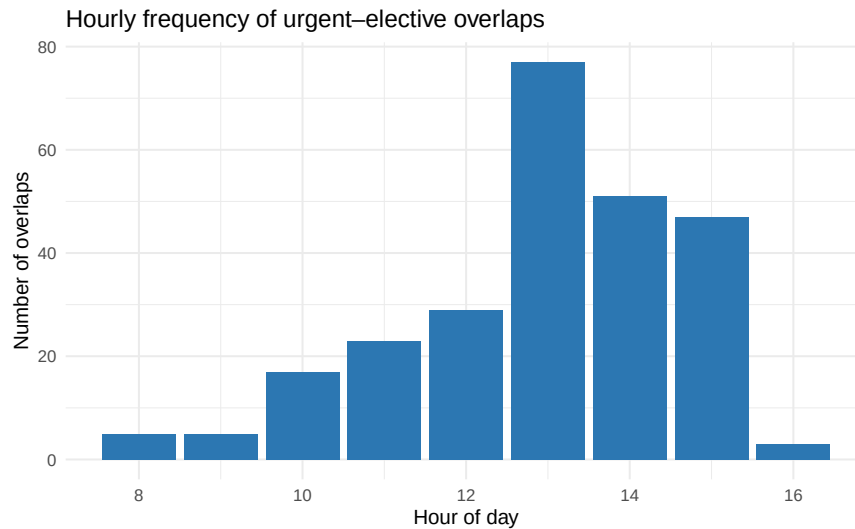


Figure 28: Hourly frequency of urgent–elective overlaps

Figure 29 shows clear differences in average start delays between elective surgeries with and without an urgent overlap. During much of 2022, the overlapped cases display strongly negative mean delays, in several months exceeding minus 50 minutes, which indicates that these surgeries tended to start significantly earlier than planned. In comparison, non-overlapped cases remained close to the planned schedule, with mean delays typically around zero to ten minutes.

From 2023 onward the gap between both groups narrows, yet overlapped cases continue to fluctuate more widely, with repeated downward spikes and intermittent positive peaks. This increased volatility suggests

that the timing of surgeries affected by urgent activity is more sensitive to local workflow pressures, leading them at times to start markedly earlier or slightly later than planned.

The recurring peaks and troughs in the overlapped series indicate that overlaps are associated with periodic scheduling tensions rather than isolated irregularities. Overall, the figure shows that urgent cases introduce measurable variability into elective scheduling, not through systematic delays but through greater unpredictability in start times, especially during periods of high operating room utilisation.

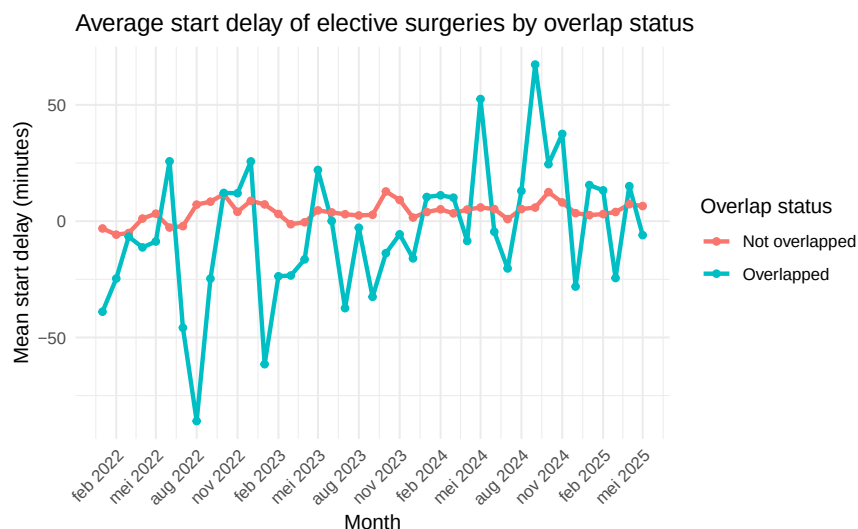


Figure 29: Average start delay of elective surgeries by overlap status

As shown in Table 39, the frequency and impact of urgent–elective overlaps vary substantially across operating rooms. The room with the highest number of overlap events is LO02, with 259 overlapped cases, representing 2.6 percent of its elective activity. This confirms that LO02 is one of the primary locations where urgent demand intersects with scheduled work. Other rooms with notable counts include LO04 (108 overlaps), LP01 (33), and LO03 (19), although their relative shares remain below 2 percent.

Median start-delay patterns show that overlaps do not affect all rooms in the same way. In LO04 and LO03, the median delay under overlap becomes negative (−16.5 and −7 minutes respectively), indicating that overlapped elective cases tend to start earlier than planned. LO01 shows a similar pattern with a median of −3 minutes. In contrast, rooms such as LP01 and LP02 exhibit positive median delays under overlap (12 and 9 minutes), suggesting that overlaps in these rooms are more likely to push elective cases later than scheduled.

A few rooms such as LO05 display extreme values (−73 minutes), but because these rooms have very small numbers of overlaps, these medians should be interpreted with caution.

Overall, the results show that the operational effect of urgent–elective overlap is concentrated in a small number of high-volume rooms. Even within these rooms, the direction of the impact is not uniform: in some settings overlaps delay elective cases, while in others they coincide with earlier-than-planned starts, likely reflecting ad hoc rescheduling, case reordering, or opportunistic use of available capacity.

Table 39: Operating rooms with urgent–elective overlap.

PlannedOR	Count	n_elective	Percentage	Med StartDelay (No Overlap)	Med StartDelay (Overlap)
LO02	259	9871	2.6	6	0
LO04	108	5806	1.9	13	-16.5

PlannedOR	Count	n_elective	Percentage	Med StartDelay (No Overlap)	Med StartDelay (Overlap)
LP01	33	18936	0.2	10	12
LO03	19	4206	0.5	5	-7
LO01	17	5200	0.3	0	-3
LP02	16	19863	0.1	-2	9
LO05	3	5223	0.1	5	-73

How stable is the surgical schedule in execution?

Beyond accurate planning, the stability of the realised schedule is a key indicator of operational performance in the operating theatre. Even when planned start times, durations, and room assignments are correct, day-to-day execution may deviate because of unforeseen delays, emergency insertions, or coordination issues.

This section evaluates how consistently the planned schedule is maintained throughout the day by focusing on two complementary aspects:

- *Shift changes*, where surgeries are executed in a different shift than originally planned, indicating how often the daily programme must be adjusted in real time.
- *Idle or gap times*, representing the pauses between consecutive surgeries within the same room, which reflect how efficiently the available capacity is utilised.

Together, these indicators capture the operational resilience of the OR complex and the ability to absorb disruptions and maintain smooth throughput despite variable circumstances. High rates of shift changes or long idle periods point to latent inefficiencies in daily planning and coordination, whereas stability across these metrics signals a well-synchronised workflow.

How often are surgeries moved to another shift?

Changes between planned and actual shifts occur when surgeries start significantly later than scheduled or are postponed to a later block of the day. The actual shift is determined based on the observed start time of each surgery relative to the official shift boundaries used at the operating department (day: 08:00–16:30, evening: 16:30–22:00, night: 22:00–08:00).

At Campus Lanaken, 578 surgeries, or 0.8 percent of all recorded cases, were performed in a different shift than originally planned, as shown in Table 40. For these surgeries, the average start delay was minus 143.1 minutes, indicating that such deviations typically involve a substantial advance within the daily schedule. Interestingly, the average deviation in procedure duration was minus 11.7 minutes, meaning that surgeries executed in another shift tend to be somewhat shorter than their planned estimates once they begin.

The mean overtime for these cases was 1 minute, showing that some shift changes result from late starts or, in other cases, procedures that exceed their planned duration, rather than extensive spillovers deep into the next shift. Although these cases represent only a small fraction of total surgical activity, their magnitude and timing can still disrupt staffing and sequencing for subsequent sessions.

Table 40: Cases performed in a different shift than planned (Lanaken).

n	MismatchCount	Share moved (%)	Mean Start Delay (min)	Mean Duration Diff (min)	Mean Overtime (min)
69395	578	0.8	-143.1	-11.7	1

As shown in Figure 30, the monthly share of surgeries performed in a different shift oscillates visibly but without any structural upward or downward trend. In early 2022, values generally range between about 0.8 and 1.3 percent, with a small number of peaks reaching roughly 1.5 percent. Throughout the remainder of the period, the pattern continues to fluctuate between approximately 0.3 and 1.3 percent, reflecting normal month-to-month variation rather than systematic change. Toward 2025, the series appears to settle into a narrower band around 0.7 to 0.9 percent.

Overall, the figure indicates that shift changes occur regularly but at a consistently low rate, and their frequency does not show signs of long-term increase or deterioration over time.

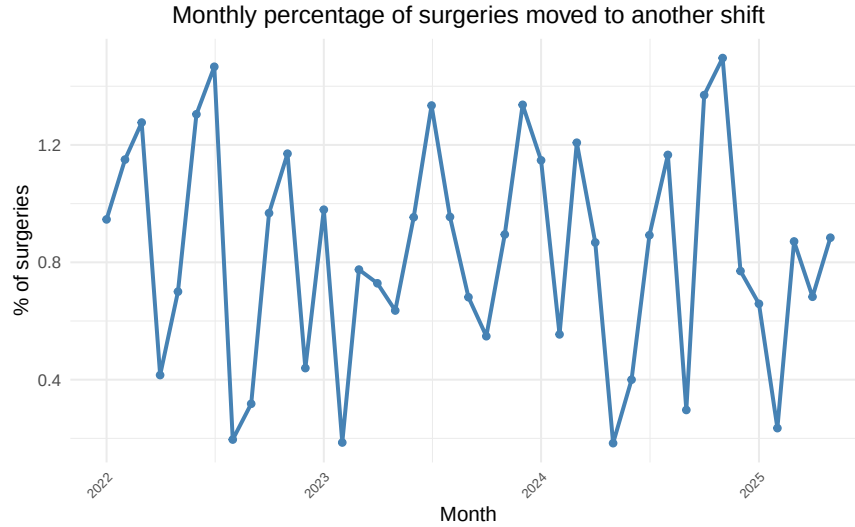


Figure 30: Monthly percentage of surgeries moved to another shift

Overall, the data suggest a modest improvement in schedule adherence after the fluctuations seen in early 2022, followed by a period of relative stability. The recurring month-to-month peaks and dips appear to reflect normal variation in workload, staffing, and case mix rather than underlying structural scheduling issues. The absence of a persistent upward trend indicates that shift changes remain a controlled and predictable feature of daily operations.

How much idle time occurs between consecutive surgeries?

Idle or gap time represents the short pause between the end of one surgery and the start of the next within the same operating room during the regular day shift (08:00–16:30). It captures how efficiently rooms are prepared for the next case and how well surgical schedules are synchronised in daily practice. For each room-day combination, the gap time measures either the delay between shift start and the first case or the interval between consecutive cases, excluding breaks longer than 60 minutes that reflect planned downtime.

At Campus Lanaken, idle time between consecutive surgeries is extremely low. As shown in Table 41, the average gap is 3.2 minutes, but the median is 0 minutes, meaning that in at least half of all transitions the next case starts immediately after the previous one ends. The interquartile range of 3 minutes confirms that most turnover intervals fall within a very narrow band, reflecting tightly sequenced workflows and minimal unused room time.

Longer gaps occur only infrequently. The 95th percentile reaches 15 minutes, and only the extreme tail approaches the cutoff of 60 minutes, which is used to exclude planned breaks rather than true idle time. Overall, the distribution indicates a highly efficient turnover process, with very limited room inactivity between cases.

Table 41: Summary statistics of idle time between consecutive surgeries (minutes).

Mean	Median	SD	IQR	Min	Max	P95	P99	P99.9
3.2	0	7.8	3	0	60	15	46	59

As shown in Figure 31, the histogram confirms that idle intervals at Campus Lanaken are overwhelmingly short. The first bar dominates the distribution: the vast majority of all transitions fall below 5 minutes,

indicating that turnover often occurs almost immediately after the previous case finishes. The next few bins show a rapid decline, with only a small fraction of intervals extending into the 5–10 or 10–15 minute range.

Beyond 15 minutes, the frequency drops sharply, and the right-hand tail becomes extremely thin. Only isolated transitions reach higher values, and gaps longer than one hour are excluded by design.

Overall, the distribution illustrates a highly efficient turnover process. Idle periods between surgeries are consistently brief, and longer gaps, while present, are rare and do not represent a systematic pattern of underutilisation.

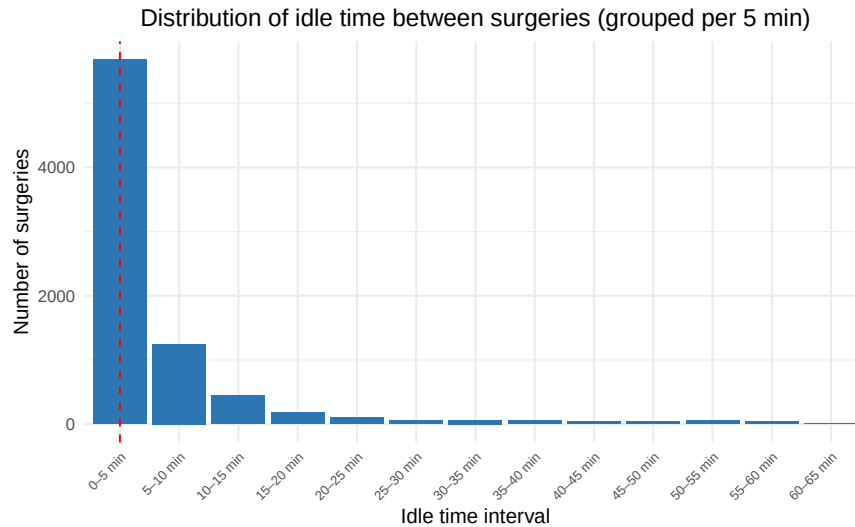


Figure 31: Distribution of idle time between surgeries (grouped per 5 min)

How does idle time vary across operating rooms?

As shown in Figure 32, idle times vary noticeably across operating rooms at Campus Lanaken. In every room, the median idle time is considerably lower than the mean, which reflects the strongly right-skewed nature of turnover intervals: most transitions occur with little to no idle time, while a smaller number of longer gaps pull the average upward.

The room-to-room differences are substantial. LO03 stands out with the highest idle times, showing both a large mean and a clearly elevated median, indicating that this room experiences longer and more frequent pauses between cases. LO05 and LO01 also show higher mean idle times compared with the rest of the OR complex, though their medians remain relatively low, suggesting occasional extended gaps rather than systematically slower turnover.

In contrast, rooms such as LP02, LP01, and especially LO02 display very short idle times, with medians at or near zero and relatively low means. These rooms appear to achieve consistently rapid turnover, likely reflecting high-throughput workflows or more predictable case sequences.

Overall, the figure shows that while turnover efficiency is generally strong across Lanaken, a few rooms exhibit noticeably longer idle intervals. These differences point to room-specific scheduling patterns or case mixes rather than systemic inefficiency.

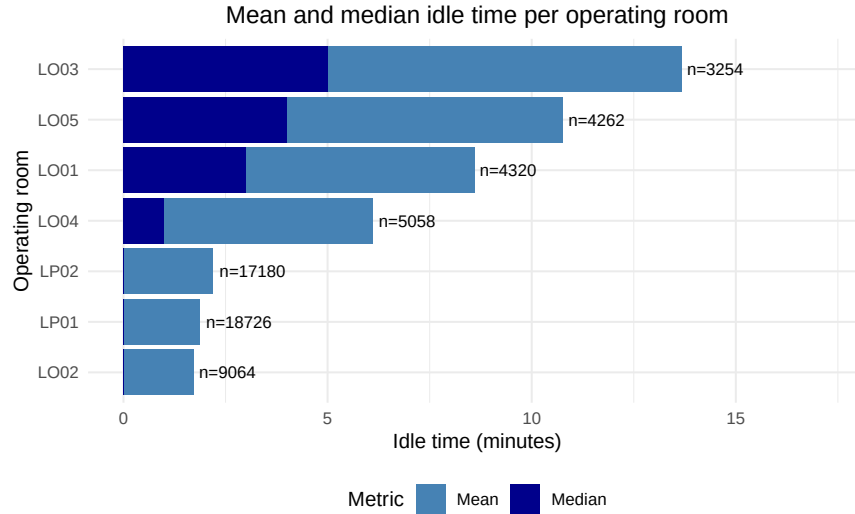


Figure 32: Mean and median idle time per operating room

As shown in Table 42, the standard deviation (SD) and coefficient of variation (CV) summarise how much idle times fluctuate around their typical value for each operating room. The SD ranges from about 5 to 11 minutes across rooms, confirming that most turnover intervals remain short even when variability increases. The CV values, however, are very high—between roughly 130 and 344 percent—because mean and median idle times in Lanaken are extremely small, causing even modest fluctuations to translate into large relative variation.

Rooms such as LO03, LO05, and LO04 stand out with the highest SDs, indicating broader spreads in turnover times and less predictable intervals between cases. These rooms experience longer gaps more frequently, consistent with what is seen in the mean–median patterns in Figure 32.

Toward the bottom of the table, rooms such as LP01, LO02, and LP02 display very high CVs despite relatively low SDs. This arises from the ratio-based nature of the CV: when typical idle times are extremely short—as they are in these rooms—even small fluctuations produce a disproportionately large coefficient of variation.

Overall, variability in idle time differs meaningfully between rooms, but the underlying distributions remain dominated by very short turnover intervals. Only a small subset of rooms shows broader spreads that may reflect more diverse schedules, irregular case sequencing, or room-specific workflow constraints.

Table 42: Standard deviation and coefficient of variation of idle time per operating room (minutes), sorted by median idle time.

ActualOR	Cases	Standard deviation	Coefficient of variation
LO03	3254	11.3	129.9
LO05	4262	9.5	141.1
LO01	4320	8.8	157.9
LO04	5058	9.8	192.4
LP02	17180	7.1	323.5
LP01	18726	6.4	343.2
LO02	9064	5.6	324.7

Conclusion

This in-depth analysis of Campus Lanaken provides a detailed and data-driven view of how surgical activity evolves from planning to execution within the operating theatre. The report systematically examines timing accuracy, room allocation, after-hours activity, urgent case dynamics, and turnover performance, offering insight into both the strengths and operational challenges of daily surgical practice.

Across the different sections, the results show that planned and observed durations are generally well aligned, though variability between procedures remains considerable. Start-time deviations are frequent but relatively contained, and their accumulation throughout the day can still lead to later-than-expected finishes. While the overall schedule performs reliably, these small deviations highlight the operational sensitivity of tightly sequenced programs.

After-hours activity is rare at Lanaken, with only about 0.5 percent of surgeries extending beyond regular hours. Most overruns are brief, yet a small number of outliers contribute disproportionately to total overtime minutes. Room assignments remain very stable, with roughly 7.5 percent of surgeries being relocated. These relocations involve mostly short, interchangeable procedures and therefore create limited disruption to the broader workflow.

Urgent cases represent a small fraction of overall activity but still create meaningful operational interaction with the elective schedule. Overlaps between urgent and elective cases occur regularly, though their impact on elective start times is modest and varies across operating rooms. In most rooms, overlapped elective cases start slightly earlier or only marginally later than planned, suggesting that urgent interventions are absorbed efficiently into the daily workflow.

The analysis of shift transitions and idle times further illustrates an efficient process. Only about 1 percent of surgeries are performed in a different shift than originally planned, indicating that schedule drift remains contained. Idle periods between surgeries are very short, with a median of 0 minutes and an average of 3.2 minutes, confirming exceptionally fast turnover performance across rooms. A few rooms display broader idle-time distributions, reflecting more varied case sequencing or temporary bottlenecks, but these represent exceptions rather than systemic issues.

Altogether, this document offers a comprehensive quantitative picture of operating room performance at Campus Lanaken. It shows how surgical planning translates into real-world execution within a largely stable and highly efficient environment. The findings confirm strong overall performance, with targeted opportunities to further stabilise start times and maintain smooth coordination between elective and urgent surgical activity.